APPLICATION FOR BOULDER COUNTY SPECIAL USE PERMIT AND SITE SPECIFIC DEVELOPMENT PLAN FOR MINING AND RECLAMATION

DOWE FLATS

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SHB AGRA Project E92-7075
June 1, 1993

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- **Baseline Air Quality and Meteorological Monitoring**
- **Noise Analysis**
- **Blasting Analysis**
- **Transportation Analysis**
- **Baseline Water Quality Monitoring**
- **Archaeology**
- **Vegetation**
- **Wetlands**
- **Wildlife**
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- **Visual Analysis**

1.0 INTRODUCTION

Pursuant to the Boulder County Zoning Resolution, Southdown, Inc. (Southwestern Portland Cement) herewith submits concurrent applications for a Special Use Permit and Site Specific Development Plan for mining and reclamation at Dowe Flats.

Approval of the permits, as submitted, would accomplish the following three goals.

1) Ensure long-term stability of the cement plant. Given historical levels of permitted output of cement products, it has been estimated that the proposed mine plan would provide cement plant reserves for 25 years.

2) Screen visual and noise impacts of the mining operation and create valuable wildlife habitat with an innovative reclamation plan. The mine will be reclaimed in a fashion compatible with the surrounding neighborhood.

3) Accommodate the construction of a reservoir in the East Valley of Dowe Flats. The mine and reclamation plan has been designed to accommodate dam construction. A separate, but concurrently processed, Special Use Application for water storage has been submitted.

The mine and reclamation plans have been designed to achieve all three of these goals during the 25 year life-of-mine.

A Regular 112 Reclamation Permit application for mining and reclamation has been made to the Colorado Mined Land Reclamation Board (CMLRB). The CMLRB application document has been submitted to Boulder County and is herein incorporated by reference into this Special Use application. Also 11 Technical Appendices are attached to this application to support the findings in this document and to comply with County submittal requirements.

Mine operations are detailed in the CMLRB permit, and this permit application should be used as a companion volume to this Boulder County Special Use Permit application, as the county permit does not duplicate all of the mine operation details found in the CMLRB permit.

Materials to be mined within the context of this plan are limestone and shale which will be

transported by haul trucks to the existing cement plant, located one mile to the south of Dowe Flats. County Road 47, in the central portion of Dowe Flats, is proposed to be relocated to accommodate mining.

Production rock will be hauled to the cement plant on a private haul road. Hauling will be done in 85 ton haul trucks crossing State Highway 66 at grade, and proceeding south into the plant. The haul road will have to be constructed prior to mining and will also cross the Burlington Northern Railroad tracks, the St. Vrain River, property owned by Frontier Materials, County Road 49 and then into the cement plant property.

Improvements to public roadways will be performed as necessary and as required by Boulder County and the Colorado Department of Transportation.

1.1 Project Summary

Included with this permit application is the "Dowe Flats Project Summary" which presents brief descriptions of:

The Proponent
Mining and Reclamation Plans
Water Storage Plans
Permits/Appraisals Required
Project Team/Contact People for Additional Information
Public and Agency Review Process
Color Photos of Existing Area, Mine Reclamation Plan, and Conceptual Reservoir
Project Vicinity Maps

1.2 Format of the Permit Application

The commentary of the Dowe Flats Mine and Reclamation Permit applications generally follows the format of Article 20, "Uses Permitted By Special Review", and Article 19, "Site Specific Development Plan, of the Boulder County Zoning Resolution (as amended February 1992)", which is the applicable regulatory document for permitting procedures within Boulder County. Although a Special Use Permit must be obtained prior to approval of a Site Specific Development Plan, they may be applied for concurrently, as in the case of this Mine and

Reclamation Permit application.

The permit commentary, generally follows each section of each Article in the Zoning Resolution. Each section of Article 20 and 19 of the Zoning Resolution is addressed, unless response to a section is considered to not apply to this proposal.

2.0 APPLICATION FORMS

Completed application forms, lists of agents, and landowner signatures are attached to this application.

3.0 PROFESSIONAL QUALIFICATIONS

The professional qualifications of the consulting team are attached to this application.

4.0 SITE PLAN

Total Acres

The total acreage within the Special Use Permit boundary is 1,911. Individual property ownership in and around the permit boundary is shown on Map 1. Map 1 also shows the mine haul road to the cement plant and the proposed corridor for the relocated County Road 47.

Location Map

Dowe Flats is located in north-central Boulder County, approximately two miles east of the Town of Lyons, five miles northwest of the City of Longmont, two miles south of Larimer County, and twelve miles north of the City of Boulder. A site location map is shown on Map 2.

Name, Address, Phone of Applicant, Engineer, Surveyor

This information is attached to this application.

Boundary Lines With Dimensions, Township & Range, Sections

The boundary of this Special Use application is shown on Map 1.

Existing and Proposed Topographic Contours

Map 1 displays existing topographic contours. Exhibit F of the CMLRB application displays the approximate post mining topographic contours.

Significant Existing and Proposed Features

Existing site features are shown on Map 3. Proposed features, including the mine and reclamation plan, are shown in Exhibits C, D, E and F of the CMLRB application.

Description of Existing and Proposed Roads, Road Rights-of-way

Existing roads are shown on Map 3. Proposed roads, including the mine haul road and the relocated County Road 47, are shown on Map 1. The main road construction activities are the relocation of County Road 47 and the construction of the private haul road. In addition to these, internal haul roads will be constructed, used temporarily, and reclaimed. These internal roads are within the mining and reclamation area.

Relocated County Road 47 will be constructed to the County's local access road standards. It will be a gravel road (as it is now) approximately 22 ft. wide (travel lanes). The mine haul road will be approximately 60 ft. wide, except for the bridge over the St. Vrain River, which will be a single travel lane 30 ft. wide.

Description of Buildings Mine Operations Center Conceptual Plan

The mine operations center for the Dowe Flats mining and reclamation project will be the Harroun farm, shown on Map 3. The Harroun farm is currently used as a feed lot and for growing crops. Although this activity is planned to continue, certain buildings will be demolished, others will be remodeled and new structures will be built. Access will remain where it currently is (off County Road 47) and parking for 20 employees and visitors will be

constructed. Lighting will be improved as required by the Mine Safety and Health Administration. The existing water and sewer on the farm will be upgraded to service mine employees.

As discussed below, 16,200 sq. ft. of existing buildings will be utilized for the mine operations center. This makes up a large part of the 25,400 sq. ft. total proposed for the operations center. Of that total, 1,700 square feet will be office related and 22,200 sq. ft. will be covered storage, mine vehicle maintenance area, and covered parking for mine operations vehicles. The 1,500 sq. ft. farmhouse at the mine operations center will be boarded up and not used for any purpose.

Another existing farmhouse, barn, and small sheds southwest of the mine operations center will continue to be used by the tenant rancher. Square footage of these are not included in the mine operations center conceptual plan.

Existing buildings and structures

Existing buildings at the proposed mine operations center site consist of a farmhouse, three animal feed stalls and associated feed lots, a metal quonset hut, three sheet metal sheds, a milk house, and various small storage sheds. The milk house, one sheet metal shed, and miscellaneous small sheds in poor condition will be removed.

Locations of existing buildings to be retained are shown on Map 3A. The approximate square footage and proposed use for each are as follows:

Farmhouse	1,500 sq. ft. - will be boarded up and not used
Metal quonset hut	7,500 sq. ft. - covered parking of haul trucks
Sheet metal shed	3,200 sq. ft. - storage and garage for pick up trucks
Sheet metal shed	4,000 sq. ft. - garage for pick up trucks
Total sq. ft. retained	16,200 sq. ft.

The three feed lots and associated feeding stalls and small grain storage sheds will be retained and continue to be used for raising farm animals.

Proposed new buildings

Locations of building envelopes for proposed new construction are shown on Map 3A.

A new single-story 1,700 sq. ft. operations center office building is proposed to be constructed with concrete block exterior treatment. This building will not exceed 35 feet in height. It will house three offices, a records room, and two bathrooms with showers to serve up to 15 men and 3 women. The office building will be located within a building envelope having the following setbacks:

North (existing Vestal Road)	1,666 feet
West (existing North 53rd Street)	748 feet
East (east Mine Permit Boundary)	4,439 feet
South (south Mine Permit Boundary)	817 feet

An equipment maintenance building may be required in the future. It is planned to be approximately 7,500 sq. ft. and constructed of prefabricated metal and concrete block. The maintenance building will not exceed 50 feet in height as required by the Boulder County Zoning Resolution Article 4-301 for Agricultural Districts. The maintenance building envelope is located with setbacks as follows:

North (existing Vestal Road)	1,913 feet
West (existing North 53rd Street)	555 feet
East (east Mine Permit Boundary)	4,483 feet
South (south Mine Permit Boundary)	529 feet

Employee Access Road and Parking

The existing access road will be retained for employee and visitor access to the mine operations center. A 6,000 sq. ft. parking area will be added adjacent to the proposed office building as shown on Map 3A. This will provide space for 20 administrative employee, mine employee and visitor cars. The access road and parking area will be used for passenger vehicles only and will be surfaced with compacted gravel.

Mining Equipment Access Road

A new 50 feet wide road providing access for mining equipment and haul trucks to the covered parking at the mine operations center is proposed. For safety reasons, this is located to keep mining equipment traffic separate from passenger vehicles. The approximate location and corridor is shown on Map 3A. The mining equipment access road will be surfaced with compacted gravel.

Lighting of the Mine Operations Center

Outdoor lighting will be installed at six locations for employee safety and will be directed so as to avoid shining off the mine operations center property. One light post will be located on the employee access road and four will be located within the proposed employee parking area. One light post will be located outside of the haul truck parking quonset hut building and directed to illuminate the interior. No lighting will exceed 25 feet in height as required by Article 22-701 of the Boulder County Zoning Resolution. General locations of light posts are shown on Map 3A.

Landscaping

Landscaping plans are fully described and depicted in the reclamation plan, Exhibits E and F of the CMLRB application.

Wells/Water Line

Water supplies associated with the existing structures on the Harroun property will be used for personnel at the mine operation center. Water required for the operation of the mine (e.g. dust control) will be supplied from the applicants' existing water rights in the area.

Design and Layout of Sewer Service

Sanitary sewer facilities associated with the existing structures on the Harroun property will be utilized for the mine operation center.

Exterior Lighting Fixtures

Lighting will be used as required by MSHA and where necessary for safety. Mining will not take place after dark.

Traffic Signs

Traffic signs will be erected as required on the relocated County Road 47. Signage will also be required for the mine haul road crossing of State Highway 66 and County Road 49 (North 51st). The applicant will work with the Colorado Department of Highways and Boulder County to determine and fulfill signage requirements.

Overlot Grading Plans

Overlot grading plans are described and depicted in the reclamation plan, Exhibits E and F of the CMLRB application.

Storm Drainage Plans

Storm drainage will be diverted and directed into the mined out areas such that no runoff from disturbed ground enters natural drainageways or ditches or will be directed through a sediment pond prior to discharge.

Parking Areas

Parking areas associated with the existing and new structures on the Harroun property will be utilized for the mine operation center.

Number of employees

The mine is expected to employ up to 20 people, some of whom are currently employed by the applicant.

5.0 DEVELOPMENT REPORT

5.1 Title Report

A title report for the property is attached to this application.

5.2 Description of Site Features

5.2.1 Geography

Dowe Flats is a triangular valley that resulted from erosion of less-resistant geologic strata in the interior of a plunging syncline. Ridges of more resistant beds surround the valley on three sides. The valley floor has low topographic relief and slopes toward the south at an average 2% gradient. There is a maximum of approximately 700 feet of topographic relief between the valley floor and the top of the ridges. The 40-foot high Limestone Ridge transects the western half of the valley. The valley is drained by small intermittent streams and ephemeral gullies.

5.2.2 Drainages

Dowe Flats is a three-sided basin, with the floor of the basin comprising some 2000 acres. Dowe Flats is so named because of the wide, flat, and uniformly-sloping floor of the valley. The area is flanked by the slopes of Rabbit Mountain to the east and Indian Mountain to the west. Dowe Pass bounds the valley on the north.

The total area of the drainage basin is comparatively small, no more than 4500 acres. A spine-like ridge rises 40 feet above the valley floor and trends due north to separate Dowe Flats into the East Valley and the West Valley (Map 3). This drainage divide, called "Limestone Ridge" (or "Hi-Cal Ridge") separates the East and West Valleys into two individual sub-basins. At approximately 3000 acres, the East Valley is the larger drainage, extending north of the West Valley to Dowe Pass. Exhibit G of the CMLRB application contains a detailed discussion of site hydrology.

The East Valley is drained by a small intermittent creek that owes some of its flow to seepage from the St. Vrain supply canal and agricultural irrigation. The West Valley lacks a well

defined drainage feature, although the valley bottom collects water in a series of wetlands, which are also augmented by seepage from the St. Vrain supply canal and agricultural irrigation. The Supply Ditch intercepts drainage from both valleys.

5.2.3 Floodways

St. Vrain Creek is the only Overlay Zoned floodplain near this project and it is not within the mine area. The only relevance to this project is the mine haul road crossing of the St. Vrain Creek Floodplain. Map 4 shows the mine haul road crossing of the Overlay Zoned Floodplain of the St. Vrain Creek. Haul road construction in the floodplain will be done in compliance with the County's Floodplain Zoning requirements.

5.2.4 High Ground Water

A detailed discussion of groundwater is contained in Exhibit G of the CMLRB application. A groundwater issue associated with this project is the Syntex landfill located north of this project in Dowe Pass. Groundwater contamination from the Syntex site has not migrated off site and does not affect the Dowe Flats project. This statement is evidenced by the following:

1. The site is under enforcement action by the State of Colorado and is undergoing active clean up.

2. The Northern Colorado Water Conservancy District (Northern), City of Boulder, and City of Longmont have signed a "Settlement Agreement" with Syntex which stipulates that the St. Vrain Water Supply Canal is not impacted by groundwater contamination from the Syntex site (the agreement is attached as an appendix to this application).

3. Northern has installed groundwater monitoring wells downgradient of the Syntex site and upgradient of the canal. The wells are sampled quarterly and indicate no contamination.

4. Baseline surface and groundwater quality monitoring for the Dowe Flats project indicates no contamination from the Syntex site is present on the Dowe Flats

project property (monitoring reports for four quarters are attached as an appendix to this application.

5.2.5 Wetlands

Existing Conditions

An inventory of federally regulated wetlands was conducted within the general confines of the Dowe Flats basin in 1987 (Aquatic and Wetland Consultants, Inc., 1987). Results of the inventory have been verified by the U.S. Army Corps of Engineers, Omaha Regulatory District, (ACOE) as documented in a letter from ACOE dated 25 September 1987. An additional wetlands inventory was conducted in 1990 (Aquatic and Wetland Consultants, Inc., 1990). Inventory results have received concurrence by ACOE (wetlands data, map and ACOE letters are attached as an appendix to this application).

A total of 45.1 acres of wetlands was inventoried in 1987, and another 8.5 acres were inventoried in 1990. The combined total of jurisdictional wetlands within the project area is approximately 53.6 acres.

Wetlands located within the valley typically follow natural drainage features or occur as isolated seeps. Some of the wetlands appear to be supported, at least in part, by seepage from the St. Vrain Supply Canal and other irrigation ditches which traverse the property. Wetlands located south of Highway 66 in proximity to St. Vrain Creek, are also supported by localized seepage, or appear to be remnant floodplain features.

Most of the wetlands within the Dowe Flats basin are dominated by herbaceous species including:

cattail	*Typha angustifolia*
three-square rush	*Scirpus americanus*
bulrush	*Scirpus acutus*
arctic rush	*Juncus arcticus*
sedges	*Carex* spp.
reed canary grass	*Phalaroides arundinacea*

Cottonwoods (*Populus* spp.) and willows (*Salix* spp.) intermittently border the small drainageways.

Riparian areas characterized by mature cottonwood and willow stands border St. Vrain Creek and the Supply, Highland, Rough and Ready, and Palmerton irrigation ditch corridors, but were not included as part of this study because they are not considered to be wetlands by federal definition. The only impacts to riparian features will be from crossings by the Mine Haul Road. There will be no mining on or near these ditches and related riparian features. These areas will be further protected by mine operation setbacks and reclamation buffers. Some ditch related riparian zones will be secluded by the reclaimed overburden features, increasing the wildlife habitat function.

Project Impacts

Mining operations in the valley will avoid wetland impact entirely with the possible exception of a temporary crossing of the unnamed drainage in the east valley. A crossing of the drainage may be required to transport overburden from the mine site to reclamation areas located east of the drainage. The temporary crossing can be authorized by a Section 404 Nationwide Permit (#26) and will be removed and reclaimed to its original condition as part of the project reclamation plan.

The Haul Road, linking the mining operation to the cement plant, will cross the St. Vrain River and one isolated wetland located in the borrow ditch of the railroad. Both the wetland and river crossing for the roadway can be authorized separately from the mining operation by Section 404 Nationwide Permits (#14 and #26).

Mitigation

Since fill in jurisdictional wetlands totaling one acre or more will be avoided, an individual Section 404 permit application for the mining and reclamation activities is not required. Notwithstanding this fact, approximately 37 acres of wetland and 44 acres of open water areas are incorporated into the mine reclamation plan design, resulting in a net increase of wetlands and aquatic resource within the valley. It is anticipated that groundwater will fill and/or saturate low-lying areas of the mine pit that will not be backfilled. These areas will develop wetland

characteristics with time and add to the diversity of habitat types that are anticipated to occur through reclamation.

5.2.6 Topography

Project topography has been previously discussed.

5.2.7 Vegetation

Existing Conditions

The vegetation of the project area has been described by Holistic Resource Management (1987), and Esco Associates, Inc. (1987). Vegetation reports for the project area are attached to this application as an appendix.

Improved Pasture

Improved pasture areas were probably once cultivated and later planted to Russian wild rye (*Elymus junceus*). These areas are heavily used by cattle and are occupied by large numbers of black-tailed prairie dogs. The cumulative effect of foraging by these animals has resulted in pronounced overgrazing and invasion of weedy plants. Invader species include:

snakeweed	*Xanthocephalum sarothrae*
musk thistle	*Carduus nutans*
Canada thistle	*Cirsium arvense*
common bindweed	*Convolvulus arvensis*
plains prickly poppy	*Argemone polyanthemos*

Hay Pasture

Hay pasture is irrigated land that has been planted to such domestic forage grasses as:

smooth brome	*Bromopsis inermis*
orchard grass	*Dactylis glomerata*

common timothy	*Phleum pratense*
tall fescue	*Festuca arundinacea*

These lands are mowed for hay during the growing season and may later be given over to grazing after the growing season.

Cultivated Land

Cultivated land is managed primarily for the production of small grains, especially winter wheat and oats, and occasionally corn. Most of this land is managed in a strip rotation pattern in which alfalfa is periodically rotated in to improve soil fertility and tilth.

Upland Grassland along Limestone Ridge

The grassland along Limestone Ridge is a remnant of the original vegetation cover on the shallow soils and convex portion of Dowe Flats. Although it has been heavily disturbed by grazing and other activities, it retains a diversity of native plants not found in the more extensive created pastures and adjacent cultivated land. Dominant species in this stand are the following perennial native grasses:

blue grama	*Bouteloua gracilis*
western wheatgrass	*Agropyron smithii*
needle-and-thread	*Stipa comata*
bottlebrush squirreltail	*Sitanion longifolium*
Fendler three-awn	*Aristida fendleriana*

along with the following perennial non-native grasses:

intermediate wheatgrass	*Agropyron intermedium*
standard crested wheatgrass	*Agropyron desertorum*
smooth brome	*Bromus inermis*
Russian wild rye	*Elymus junceus*

Native annual grasses present in very small amounts are:

sixweeks fescue	*Vulpia octoflora*
false buffalograss	*Munroa squarrosa*

Introduced annual grasses are locally abundant, and include:

cheatgrass (especially abundant)	*Bromus tectorum*
Japanese brome	*Bromus japonicus*
annual ryegrass	*Secale cereale*
wheat	*Triticum aestivum*

Shrubs are scattered sparsely but rather uniformly and include:

rubber rabbitbrush	*Chrysothamnus nauseosus*
winterfat	*Krascheninnikovia lanata*
three-leaf sumac	*Rhus trilobata*
prickly rose	*Rosa acicularis*
snakeweed	*Xanthocephalum sarothrae*

Among the numerous native perennial forbs present are:

textile onion	*Allium textile*
prickly poppy	*Argemone polyanthemos*
fringed sage	*Artemisia frigida)*
white prairieaster	*Aster ericoides*
Porter aster	*Aster porteri*
slender milkvetch	*Astragalus gracilis*
three-leaf milkvetch	*Astragalus tridactylicus*
downy paintbrush	*Castilleja sessiliflora*
mouse-ear chickweed	*Cerastium arvense*
James hiddenflower	*Cryptantha jamesii*
wafer-parsnip	*Cymopterus acaulis*
white prairie clover	*Dalea candida*

low fleabane	*Erigeron pumilus*
slenderbush wildbuckwheat	*Eriogonum effusum*
robust spurge	*Euphorbia robusta*
wallflower	*Erysimum asperum*
evolvulus	*Evolvulus nuttalliana*
blanketflower	*Gaillardia aristata*
scarlet gaura	*Gaura coccinea*
ironplant goldenweed	*Haplopappus spinulosus*
stiff sunflower	*Helianthus ridgidus var. subrhomboideus*
hairy goldenaster	*Heterotheca villosa*
thread-leaved hymenopappus	*Hymenopappus filifolius*
low bladderpod	*Lesquerella montana*
sandlily	*Leucocrinum montanum*
spotted gayfeather	*Liatris punctata*
Lewis flax	*Linum lewisii*
salt-and-pepper	*Lomatium orientale*
bractless blazingstar	*Mentzelia nuda*
yellow blazingstar	*Mentzelia speciosa*
lanceleaf chimingbells	*Mertensia lanceolata*
musineon	*Musineon divaricatum*
false dandelion	*Nothocalais cuspidata*
yellow stemless eve. primrose	*Oenothera brachycarpa*
cutleaf evening primrose	*Oenothera coronopifolia*
Colorado loco	*Oxytropis lambertii*
James nailwort	*Paronychia jamesii*
sidebells penstemon	*Penstemon secundiflorus*
plains bahia	*Picradeniopsis oppositifolia*
narrowleaf scurfpea	*Psoralea tenuiflora*
Britton skullcap	*Scutellaria brittonii*
lambstongue groundsel	*Senecio integerrimus*
broom groundsel	*Senecio spartioides*
stiff goldenrod	*Solidago rigida*
cowboy's delight	*Sphaeralcea coccinea*
Rio Grande greenthread	*Thelesperma megapotamicum*

Easter daisy	*Townsendia cf. hookeri)*
Dakota vervain	*Verbena ambrosifolia*

Introduced perennial forbs present in the grassland on Limestone Ridge include:

field bindweed	*Convolvulus arvensis*
black medic	*Medicago lupulina*
alfalfa	*Medicago sativa*
common dandelion	*Taraxacum officinale*

Native annual and biennial forbs present include:

thyme-leaf spurge	*Chaemaesyce serpyllifolia*
gray goosefoot	*Chenopodium incanum*
yellow-center thistle	*Cirsium ochrocentrum*
wavy-leaf thistle	*Cirsium undulatum*
hare's ear	*Conringia orientalis*
Carolina draba	*Draba reptans*
Ellisia	*Ellisia nyctelea*
snow-on-the-mountain	*Euphorbia marginata*
plains sunflower	*Helianthus petiolaris*
Virginia pepperweed	*Lepidium virginicum*
dragonhead	*Moldavica parviflora*
black nightshade	*Solanum americanum*

A variety of introduced annual and biennial forbs and grasses are abundant in this area, including:

alyssum	*Alyssum minus*
redroot pigweed	*Amaranthus retroflexus*
littlepod falseflax	*Camelina microcarpa*
musk thistle	*Carduus nutans*
diffuse knapweed	*Centaurea diffusa*
purple mustard	*Chorispora tenella*

pinnate tansymustard	*Descurainia pinnata*
flixweed tansymustard	*Descurainia sophia*
filaree	*Erodium cicutarium*
common sunflower	*Helianthus annuus*
summercypress	*Kochia iranica*
stickseed	*Lappula redowskii*
false salsify	*Podospermum laciniatum*
Russian-thistle	*Salsola iberica*
Jim Hill mustard	*Sisymbrium altisissimum*
salsify	*Tragopogon dubius*
bracted verbena	*Verbena bracteata*

Lowland Grassland

These areas occur on heavy soils on low slope areas in the overburden storage area on the northern end of the West Valley of Dowe Flats and downslope from the St. Vrain Supply Canal. The dominant species present are the following perennial native grasses in order of dominance:

western wheatgrass	*Agropyron smithii*
streambank wheatgrass	*Agropyron dasystachyum* var. *riparium*

Other species locally conspicuous include the perennial grass foxtail barley (*Hordeum jubatum)*, the annual grasses cheatgrass (*Bromus tectorum*) and Japanese brome (*Bromus japonicus*), and the following perennial forbs:

fog fruit	*Phyla cuneifolia*
two-groove milkvetch	*Astragalus bisulcatus*
Short milkvetch	*Astragalus shortianus*
Dalmation toadflax	*Linaria dalmatica*
Pacific aster	*Aster adscendens*
bastard toadflax	*Comandra umbellata*
cowboy's delight	*Sphaeralcea coccinea*

Also common are annual (or biennial) forbs including:

pennyroyal	*Hedeoma hispida*
alyssum	*Alyssum minus*
moth mullein	*Verbascum blattaria*

Other Vegetation Types of the Area Not to be Disturbed by Direct Effects of Mining

Vegetation of the permit area that will not be disturbed by direct effects of mining, but which could be disturbed by ancillary facilities includes:

Natural vegetation types

Upland Grassland with Shrubs
Mountain Mahogany/Skunkbrush Shrubland
Ponderosa Pine Woodland
Riparian Shrubs in Upland Drainages
Riparian Trees and Shrubs along Irrigation Ditches
Riparian Forest
Skunkbrush/Plum/Chokecherry Shrubland
Wetland

Land use-defined vegetation types

Dry Pasture

Upland Grassland with Scattered Dryland Shrubs

This grassland unit is characterized by the occasional occurrence of mountain mahogany (*Cercocarpus montanus*) and skunkbrush (*Rhus trilobata*) shrubs, mainly on microsites of shallow soil, where grass competition is limited. The grass matrix is dominated by a mixture of cool and warm season native perennial species including:

western wheatgrass	*Agropyron smithii*
needle-and-thread	*Stipa comata*
junegrass	*Koeleria macrantha*
sideoats grama	*Bouteloua curtipendula*

blue grama	*Bouteloua gracilis*
little bluestem	*Schizachyrium scoparium*
Indiangrass	*Sorghastrum nutans*

Some local areas are overwhelmingly dominated by the introduced annual grasses Japanese brome (*Bromus japonicus*) and cheatgrass (*Bromus tectorum*).

Mountain Mahogany/ Skunkbrush Shrubland

An extensive area is distinguished by a heavy cover of mountain mahogany. In some parts, there are scattered ponderosa pine (*Pinus ponderosa*). Herbaceous understory consists of sparse cover of species mentioned in the next type description.

Ponderosa Pine Woodland

This vegetation type is dominated by ponderosa pine which provides from about 30 to 100 percent canopy cover. The understory is variously comprised of native grasses and shrubs. Ponderosa Pine Woodland grades into Mountain Mahogany Shrubland, and thus may have a dense mountain mahogany shrub understory.

Riparian Shrubs and Small Trees along Upland Drainages

Occurring in narrow strips along a few sheltered drainages are developments of mesic shrubs including:

American plum	*Prunus americana*
chokecherry	*Padus virginianus* var. *melanocarpa*
willow	*Salix* sp.
hawthorn	*Crataegus* sp.

The willows and hawthorns reach heights of a few meters.

Tree species occurring sporadically include:

boxelder	*Acer negundo*
plains cottonwood	*Populus deltoides*
lanceleaf cottonwood	*P.* x *acuminata*
peachleaf willow	*Salix amygdaloides*

Riparian Trees and Shrubs Along Irrigation Ditches

The tree species plains cottonwood and peachleaf willow intermittently border the four irrigation ditches present in the south end of Dowe Flats. Occurring with these trees are shrubs including:

Russian olive	*Eleagnus angustifolia*
sandbar willow	*Salix exigua*
hawthorn	*Crataegus* sp.

Riparian Forest along Major Watercourses

This type is confined to the immediate proximity of South St. Vrain Creek where the proposed transportation corridor crosses the creek. Dominant trees present are:

peachleaf willow	*Salix amygdaloides*
crack willow	*Salix fragilis*
boxelder	*Acer negundo*
plains cottonwood	*Populus deltoides*
narrowleaf cottonwood	*Populus angustifolia*

Other tree species present include:

Russian olive	*Eleagnus angustifolia*
sandbar willow	*Salix exigua*
New Mexico locust	*Robinia neomexicana*

Shrub species present are:

prickly rose	*Rosa acicularis*
Wood's rose	*Rosa woodsii*
chokecherry	*Padus virginiana* var. *melanocarpa*
wild plum	*Prunus americana*
snowberry	*Symphoricarpos* cf. *occidentalis*

Perennial grasses and grasslikes present include:

mannagrass	*Glyceria striata*
Kentucky bluegrass	*Poa agassizensis*
reed canarygrass	*Phalaris arundincaea*
Nebraska sedge	*Carex nebrascensis*
broom sedge	*Carex scoparia*
foxtail	*Alopecurus pratensis*

Perennial forbs present include:

spotted knapweed	*Centaurea maculata*
American speedwell	*Veronica americana*
willowherb	*Epilobium adenocaulon*
bouncing bet	*Saponaria officinalis*
field mint	*Mentha arvensis*
Canada thistle	*Cirsium arvense*
cress	*Rorippa palustris* ssp. *hispida*

Annual (and biennial) forbs present include:

ragweed	*Ambrosia* sp.
musk thistle	*Carduus nutans*
burdock	*Arctium minus*

Vines present are:

western virgin's bower	*Clematis ligusticifolia*
wild hops	*Humulus lupulus*

Skunkbrush/ Plum/Chokecherry Shrubland

In these areas, deep colluvial deposits support thickets of the following shrubs:

skunkbrush	*Rhus trilobata*
American plum	*Prunus americana*
Chokecherry	*Padus virginiana* var. *melanocarpa*
squaw currant (occasional)	*Ribes cereum*
hackberry (occasional)	*Celtis occidentalis*

The American plum and chokecherry do not achieve their typical heights, staying mostly below one meter. This is attributable to the fact that they occur on south- and west-facing slopes that are naturally relatively dry and hot.

Wetland

These areas, occurring in depressions and drainage bottoms, qualify as wetlands due to the abundance of wetland indicator species at particular locations and may include portions of the Riparian Forest unit. Species include:

Nebraska sedge	*Carex nebrascensis*
broom sedge	*Carex scoparia*
Baltic rush	*Juncus arcticus* ssp. *balticus*
interior rush	*Juncus interior*
threesquare	*Scirpus americanus*
foxtail barley	*Hordeum jubatum*
tall fescue	*Festuca arundinacea*
curly dock	*Rumex crispus*

In the wettest areas, broadleaf cattail (*Typha latifolia*) and hardstem bulrush (*Scirpus acutus*) occur. There are local areas of heavy cover by reed canarygrass (*Phalaris arundinacea*).

Dry Livestock Pasture

This type consists of areas with various histories, united by the presence of dry, well-drained soils and periodic heavy livestock grazing disturbance. The dominant species reflect these conditions and include the following perennial grasses and grasslikes:

smooth brome	*Bromopsis inermis*
Kentucky bluegrass	*Poa agassizensis*
sun sedge	*Carex heliophila*

the annual grass cheatgrass (*Bromus tectorum*),

several perennial forbs:

hairy goldenaster	*Heterotheca villosa*
gumweed	*Grindelia squarrosa*
alfalfa	*Medicago sativa*
spotted knapweed	*Centaurea maculata*
common bindweed	*Convolvulus arvensis*

the following annual (or biennial) forbs:

mullein	*Verbascum thapsus*
musk thistle	*Carduus nutans*

the following shrubs:

rubber rabbitbrush	*Chrysothamnus nauseosus*
Wood's rose	*Rosa woodsii*

and the succulent pricklypear cactus (*Opuntia polyacantha*).

Other Vegetation Mapping Units

In addition to the above-described vegetation types, three miscellaneous land use areas have been delineated.

Commercial property

These are areas where the surface is occupied by business facilities.

Non-agricultural Disturbed Land

This category includes various mostly linear areas disturbed by public roads, the St. Vrain Supply Canal, and irrigation ditches on the south end of Dowe Flats.

Residential Property

These are areas occupied by private residences, mostly in close proximity to Highway 66 south of Dowe Flats.

Project Impacts

Vegetation to be impacted by the proposed Dowe Flats mining operation is predominantly of an agricultural nature and is either currently under cultivation or consists of previously planted European or Asian forage species. Any remaining native vegetational cover is situated on the crest and upper slopes of Limestone Ridge and the northern portion of the West Valley. Although the area has been heavily impacted by grazing and other agricultural activities, it retains a substantial diversity of native plants not found in the more extensive created pastures and adjacent cultivated land. Numerous field surveys have indicated that no species considered rare, threatened or endangered presently exist on Limestone Ridge. A file search of the Colorado Natural Areas Program plant inventory list revealed that no plant species of concern are present in the project area. A survey for Spiranthes Diluvialis will be performed in July or August, 1993 and the results will be forwarded to Boulder County.

During the mining activity phase of the project, the vegetation of the quarry sites will be

removed. After regrading and reclamation, the restored vegetation will comprise far more diverse biological communities than the present ones, which are limited by homogeneous, nearly flat topography and biological simplification caused by agricultural use (refer to Exhibits E and F of the CMLRB application).

Mitigation

During the reclamation phase of the project, existing degraded vegetational and soil resources will be replaced by a landscape providing a greater diversity of topography, vegetation, and wildlife habitat. Aesthetic qualities will also be enhanced. Vegetation types will include upland grasslands and shrublands, wet meadows and marshes, as well as open water, and moist shrubland and forests in protected coves.

5.3 Soils

Main Soil Types of the Permit Area

Soils of the Dowe Flats project area were mapped by the U.S. Soil Conservation Service (1975) as part of the Boulder County Soil Survey (see also Exhibit I of the CMLRB application). The three main units mapped in areas that will be disturbed by mining are LaPorte very fine sandy loam, Manvel loam, and Gaynor silty clay loam.

LaPorte Soils

The LaPorte soils are shallow and well-drained and develop over limestone and limy shale. They are of moderate permeability, but because of their shallowness, have low water-retention capacity.

The upper horizon on the Dowe Flats site is a sandy loam from 4 to 7 inches thick, which is somewhat thinner than the 8 inch thickness described in SCS (1975) for this horizon. Location of LaPorte soils in this area coincides closely with the 1975 SCS data, although some areas mapped by SCS as Gaynor are actually LaPorte. Thickness of the A horizon varies from 4 to 10 inches.

Manvel Soils

The Manvel soils are deep and develop on calcareous, loamy alluvium, which is located on the broad central areas of Dowe Flats, away from limestone outcrops and the slopes of Rabbit Mountain. The thickness of the surface (A) horizon varies from 4 to 6 inches.

Gaynor Soils

The Gaynor soils are moderately deep soils derived from loamy alluvial and wind-blown materials. They are located in areas of wind deposits, sites difficult to accurately predict. Many of the areas mapped by the SCS as Gaynor are actually of the LaPorte or Manvel series. The surface material likely to be salvaged for reclamation (A and AC horizons) averages 8 inches in thickness.

Other Soils of the Permit Area

Soil units that will not be disturbed by the Dowe Flats project or which will experience only very minor disturbance include: Ascalon sandy loam, Ascalon-Otero complex, Baller stony sandy loam, Calkins sandy loam, Loveland soils, Nunn sandy clay loam, Nunn clay loam, Valmont clay loam, and Valmont cobbly clay loam.

Ascalon Soils

Ascalon soils are fine loamy, mixed, mesic Aridic Argiustolls, deep and well-drained, formed on terraces and uplands in loamy mixed alluvium and wind-laid materials. A sandy loam surface horizon of about 8 inches is underlain by a textural B-horizon of about 11 inches. Below this, a calcareous subsoil extends to a depth of 60 inches or more.

Baller Soils

Baller soils are loamy-skeletal, mixed, xeric Lithic Haplustolls, shallow and well-drained, and are formed on upland ridges and slopes in loamy residuum weathered from sandstone. Typically, the surface layer is a grayish brown stony sandy loam about 10 inches thick.

Below this, 5 inches of light brownish gray very stony sandy loam overlies the sandstone parent material.

Calkins Soils

Calkins soils are coarse loamy, mixed, mesic Cumulic Haplaquolls, are deep and somewhat poorly drained, and are formed from loamy alluvium on low terraces and bottomlands. Typically, the surface layer is a grayish brown sandy loam extending to a depth of 40 inches. Below this, a light-brownish gray coarse sandy loam extends to 60 inches or more.

Loveland Soils

Loveland soils are fine loamy over sandy or sandy-skeletal, mixed (calcareous), mesic Typic Haplaquolls, are deep and somewhat poorly drained, and are formed on loamy alluvium overlying sandy and gravelly material on terraces and bottomlands. The surface layer is typically a calcareous dark grayish brown light clay loam about 20 inches thick, mottled in the lower parts. Below this a layer of gypsic and strongly calcareous mottled light clay loam about 10 inches thick overlies mottled light brownish-gray gravelly sand extending to a depth of 60 inches or more.

Nunn Soils

Nunn soils are fine montmorillonitic, mesic Aridic Argiustolls, are deep and well-drained, and are formed on terraces and valley side slopes in loamy alluvium. Typically the surface layer is a grayish brown clay loam about 10 inches thick. The subsoil is a brown and very pale brown clay, grading to clay loam, about 10 inches thick. Below this, the substratum is a calcareous very pale brown clay extending to 60 inches or more.

Otero Soils

Otero soils are coarse loamy, mixed (calcareous), mesic Ustic Torriorthents, are deep and well-drained, and are formed on terraces, rolling uplands, and valley slopes in loamy alluvium and wind-laid materials. Typically, the surface layer is a strongly calcareous brown sandy loam about 6 inches thick. The subsoil is a pale brown sandy loam about 6 inches thick. The

substratum material is a strongly calcareous light yellowish brown sandy loam, extending to a depth of 60 inches or more, with lime concretions in the upper parts.

Valmont Soils

Valmont soils are clayey over loamy-skeletal, montmorillonitic, mesic Aridic Argiustolls, are deep and well-drained, and are formed on old, high terraces in cobbly and gravelly loamy alluvium. The surface layer is a grayish brown light clay loam, with varying amounts of gravel and cobbles, about 4 inches thick. An upper subsoil layer of brown clay loam about 3 inches thick overlies a middle subsoil layer of brown light clay about 13 inches thick. The lower subsoil is a calcareous light brown gravelly clay loam about 4 inches thick. Below this, a layer of pinkish-white and light brown very gravelly loam extends to a depth of 60 inches or more.

Soils Suitability

Soil samples have been submitted to the Colorado State University Soils Laboratory for verification of reclamation suitability. Slight modifications to the boundaries of soil units have been made, but are not considered to be critical because all material is anticipated to be rated as suitable for revegetation, and all units are likely to be salvageable to similar depths.

Based on recommendations by the USDA Soil Conservation Service (SCS), soils at this site are not suitable for cultivation due to erosion hazard and slow permeability. Soils on the project site have been cultivated historically, and have suffered some degradation of surface horizon thickness as a consequence. The reclamation plan is based upon SCS recommendations which indicate that the highest use of these soils would be that of supporting a cover of perennial vegetation to be used for livestock grazing and/or wildlife habitat (see also Holistic Resource Management, 1987).

During mining these soils will be sequentially removed, stored temporarily, and relocated, then revegetated to a perennial vegetational cover. The existing soil profiles will be lost during salvage, but after reclamation the soil will be overlain by a protective, mostly native

vegetational cover that will allow soil development to proceed to a degree that it has been unable to reach during its most recent history of agricultural use (refer to Exhibit I of the CMLRB application).

5.4 Geological Characteristics

5.4.1 Background

In 1985 a study was completed by Fox Consultants that provided a detailed summary of the geologic setting of the Dowe Flats area for the purpose of assisting in the planning and execution of subsurface explorations and engineering investigations in this area. This study addresses regional and local bedrock stratigraphy and structural geology. The scope of the Fox report has included compilation of existing geologic data through August 1984 and interpretation of these data aided by aerial photography interpretation and site reconnaissance. Map 5 shows a geologic map of the project area.

5.4.2 Regional Geologic Setting

Regional Stratigraphy

A maximum of approximately 12,700 feet of sediments is found in the vicinity of Dowe Flats. Only those formations beneath the lower Pierre Shale are present at Dowe Flats. In general, the geologic section is composed of alternating sandstone and shale layers with some limestones. Beginning at the bottom of the section and working upward in both depth and time, the individual units display a variety of depositional origins

Precambrian basement rock, including both plutonic and metamorphic rocks, is exposed in the mountains west of Lyons. These rocks are generally massive, hard, and tough when fresh, but coarser-grained materials can weather to grus. The basal Fountain Formation was deposited during erosion of the ancestral Rocky Mountains. The Lyons Formation above it is a beach deposit. The overlying Lykins Formation is a marine shale and the Entrada is an eolian deposit. The Morrison Formation and the Lytle Formation of the Dakota Group are both continental deposits, while all units between the South Platte Formation of the Dakota Group and the Pierre Shale were deposited in various marine environments as the seas transgressed and receded

through the area. The overlying Fox Hills Sandstone is a transitional beach type sandstone formed as the seas regressed from the area for the last time, and the uppermost Laramie Formation is a series of interbedded nonmarine claystones and shales with an occasional coal seam. During the Tertiary period, Precambrian, Paleozoic and Mesozoic rocks were invaded by igneous intrusives. One such large intrusive body is present about 2 miles southwest of Lyons and is currently being developed for commercial aggregate and riprap.

Mineral utilization near Dowe Flats includes production of cement from the Fort Hays limestone in a pit south of Dowe Flats, and quarrying of Lyons sandstone west of the site for building blocks. Existing and former gravel pits are present along St. Vrain Creek just south of Dowe Flats from Lyons to Longmont. Farther to the east are several oil fields that produce from sandstones and limestones at depth. Coal mining occurred in the early part of the century from the Laramie Formation, but is presently inactive in the area.

Regional Geologic Structure

The Dowe Flats area is in the Foothills Belt, a transition zone between the Front Range physiographic province to the west and the Denver Basin physiographic province to the east. The Front Range, which is the eastern most range of the Southern Rocky Mountains, begins on the northern side of the Arkansas River in southern Colorado and extends northward for approximately 185 miles to the Wyoming border. The range varies from 25 to 45 miles in width (Boos and Boos, 1957), and was formed by vertical uplift and subsequent erosion of the sedimentary strata to expose the Precambrian core. Remnants of the original sedimentary cover, now present as truncated sedimentary rocks along the uplift flank, are tilted by drag along the uplift boundaries. The more resistant tilted rocks, generally sandstones and some carbonates, form linear hogbacks that parallel the mountain front. The less resistant shales have been eroded away to form linear valleys between the hogbacks. This linear system of valleys and ridges is present along most of the eastern Front Range boundary.

The Denver Basin is a major structural basin east of the Rocky Mountains beginning south of Pueblo, Colorado and extending northward to the vicinity of Torrington, Wyoming. It spans approximately 180 miles at its widest point just south of the Colorado border. The basin is highly asymmetric with an axis that is subparallel to the mountain front and about 10 to 30 miles east of it. The proximity of the axis to the mountains results in a steep western limb and a

gently sloped eastern limb.

The Foothills Belt is a transitional area about 5 to 10 miles wide between these two major physiographic provinces. The sedimentary beds adjacent to the Precambrian mountain front are steeply dipping and occasionally overturned. The dip progressively decreases in the younger formations as they outcrop at greater distances from the mountain front. Between Lyons and the northern Colorado border a series of northwest-trending high-angle bedrock faults offset the sedimentary beds. Draping of sediments over these faults produces a series of en-echelon folds and faults. Taken together, these intermediate-scale structural deformations, regional deformations produced by the Front Range and the Denver Basin, and more localized small-scale folding and faulting, result in a generally complex geologic setting throughout the Foothills Belt.

5.4.3 Dowe Flats Site Geology

Local Stratigraphy

The important formations in this area are all Upper Cretaceous in age and are limited to the middle of the regional stratigraphic column. They include the Dakota Group, the Benton Formation, the Niobrara Formation and the Pierre Shale. The general stratigraphic descriptions in this section are summarized from three graduate theses that contained detailed lithologic descriptions compiled during field mapping and section measurement activities (Quam, 1932; Hunter, 1947; Masters, 1957). These documents provide the best available specific lithologic information on Dowe Flats. The most important stratigraphic units in Dowe Flats are, from oldest to youngest, the Dakota Group, containing the Lytle and South Platte formations; the Benton Formation; the Niobrara Formation, containing the Fort Hays and Smoky Hill members, and the Pierre Shale.

Dakota Group - The Dakota Group contains the upper beds on the cuestas surrounding Dowe Flats. It has an average thickness in this area of 330 feet, and is subdivided into the lower Lytle and upper South Platte formations.

The Lytle Formation consists of nonmarine fluvial deposits. The lowest part is a fine to coarse-grained massive brown sandstone intercalated with a basal conglomerate

containing quarter to half-inch diameter chert pebbles and granite fragments mixed with finer materials in a secondary silica cement. This bed is approximately 40 feet thick. The upper portion of the Lytle Formation consists of a series of variegated red and yellow claystones that, on exposure, weather to a reddish surface soil. This deposit varies in thickness from 30 to 60 feet.

The South Platte Formation constitutes the upper part of the Dakota Group. Deposits in this interval are marine and near-marine in origin. The Plainview Member is a platy, fine-grained, hematite-stained quartzose and sandstone. It varies in thickness from 20 to 30 feet through differential incising of the underlying claystones. The middle part of the South Platte Formation is a 125 to 175-foot thick gray to black carbonaceous shale interbedded with thin, buff-colored siltstones and sandstones. The uppermost part of this formation is the Muddy member, a massive, ridge-forming tan quartzose sandstone 20 to 30 feet thick that is slightly cross-bedded and distinctly jointed.

The Benton Formation - The Benton Formation is a 500-foot thick layer of fine-grained marine deposits. The lowest part is composed of dark gray to black fossiliferous sandy shales. The middle portion is a zone of light to dark gray argillaceous limestone and dark gray to black calcareous shales. The upper part is calcareous dark gray sandy shales. Numerous thin but laterally continuous bentonite layers are found throughout the formation. The lower, middle, and upper parts are often referred to as the Graneros Shale, the Greenhorn Limestone, and the Carlile Shale, respectively.

The Codell Sandstone is the uppermost unit in the Benton Formation. In the Dowe Flats area, it has a 15-foot total thickness and can be divided into a 7-foot thick gray siltstone beneath an 8-foot thick silty sandstone. This silty sandstone directly underlies the Niobrara Formation.

The Niobrara Formation - The Niobrara Formation is traditionally separated into two units, the Fort Hays and Smoky Hill members. The Fort Hays Member is an extremely fine-grained, light gray limestone with thin interbedded shales. A section of Fort Hays Member at the south end of Dowe Flats measured 16.5 feet thick (Lowman, 1977). Limestone accounted for 13.6 feet or 82% of the outcrop thickness. The limestone is distributed as blocks ranging in thickness from 0.5 to 3.1 feet and vertical joints spaced on 1 to 3-foot centers. The remaining 2.9 feet of material is distributed as 11 thin bentonite layers having an average thickness of 6 inches. Drilling in other areas within the Dowe Flats valley has indicated an average limestone thickness of 20 feet (Mallette, 1962).

The overlying Smoky Hill Member of the Niobrara Formation is generally described as a dark gray, calcareous, fossiliferous marine shale. However, characterization of the Smoky Hill Member as shale on a regional scale does not account for several separate limestone beds present in the Dowe Flats vicinity. At Dowe Flats, limestones within the Smoky Hill Member have been described at an outcrop along the Little Thompson River (Quam, 1932) and mapped in the Dowe Flats valley (Mallette, 1962). An 11-foot thick limestone unit 100 feet above the base of the Smoky Hill Member was mapped by Mallette in Dowe Flats, but was not described by Quam in the Little Thompson River outcrop. All the other materials in the bottom 200 feet of the Smoky Hill Member are dark gray to black, pyritiferous, calcareous, marine shales. A second 20-foot thick limestone bed was mapped at Dowe Flats and measured in the Little Thompson outcrop at the interval from 200 to approximately 220 feet above the bottom of the Smoky Hill. A final limestone bed, with a base about 256 feet above the Smoky Hill Member, was also located in both field efforts; however, its thickness was measured as 12 feet by Mallette (1962) and 43 feet by Quam (1932). Based upon interpretation of geophysical logs from oil exploration wells drilled 10 to 15 miles south and east of the site (Lowman, 1977), the 43-foot value appears to be more realistic.

The fact that the limestones in the Smoky Hill Member are not discussed in regional geologic summary papers indicates that they are local in nature and probably disappear to the north and south.

The Pierre Shale - The lower Pierre Shale is the uppermost unit considered in this

report. Only the lower 500 feet of the Pierre Shale are found in Dowe Flats; the remainder has been eroded away. The lower 2500 feet of Pierre Shale are homogeneous dark brown to gray-black marine shale that weathers to a buff color. The basal portion of the Pierre Shale, immediately above the Niobrara Formation, is sandy, but the sand content decreases in the main portion of the shale.

Local Structure

This section briefly introduces the major structural components of the Dowe Flats area.

The area surrounding Dowe Flats is structurally complex with several types of structural features of differing scales overprinted upon each other. Despite detailed studies, interpretations, and reinterpretations, the origin and exact relationship between all of the structural features remains unclear.

The regional Dakota hogback that parallels the eastern side of the Front Range through central and northern Colorado forms the ridge that separates Dowe Flats from the town of Lyons. The hogback ridge is normally a monocline formed by the uplift of the Front Range, but in this area it appears as the eastern limb of a doubly-plunging anticline with
an axis approximately located on the eastern edge of the Lyons town limits.

On the northern side of Dowe Flats in the vicinity of Dowe Pass, the regional hogback has been offset approximately 2 miles by a major northwest-trending high-angle fault with a trace along the Little Thompson River. This fault is one of the many large faults that have offset the hogback. The southwest side containing Dowe Flats is the downthrown block, with displacement estimated at 600 feet (Hunter, 1947).

Rabbit Mountain is a large, southward-plunging anticline that forms the ridge east of Dowe Flats. It has a steeply-dipping eastern limb and a gently-dipping western limb, a structural style opposite to the general trend of folding along the Front Range. In addition, several smaller faults and canoe folds overprinted on this anticline distort and disrupt the bedding to produce further structural complexity.

Dowe Flats itself is underlain by a southward-plunging syncline. This syncline is nearly

symmetrical with no significant folding and faulting at the south end of Dowe Flats. Its northern boundary is the extremely complex Dowe Pass area, an intensely folded area containing numerous anticlines and synclines along with some faulting.

Two faults are shown within the eastern portion of Dowe Flats by Mallette (1962). One high-angle northeast-trending fault is located in the northwestern quadrant of Section 15. The other fault is located in the eastern half of Section 15 and trends northwest. Both faults are inferred and are shown with very small offsets. However, documentation of evidence for these faults is not provided by Mallette's report, and they were not mapped by the several other major investigators of the Dowe Flats area. In addition, examination of large-scale aerial photography in connection with this study did not reveal any evidence for the northeast-trending fault and only limited evidence for the northwest-trending fault. Therefore, the northeast-trending fault is not believed to be present, and the northwest-trending fault, if present, is only a localized small-scale feature.

All of these units are upper Cretaceous in age. The synclinal structure of the central Dowe Flats basin is such that the youngest (Pierre Shale) formation is exposed over much of the eastern portion of the valley bottom with the older strata forming concentric horseshoe-like rings as they outcrop around the valley perimeter on the east and west margins.

Structurally, the region surrounding Dowe Flats is extremely complex with both intense and sometimes superimposed folding as well as Post-Cretaceous faulting. In this context, the simplicity of the central interior portion of Dowe Flats is almost anomalous. Interpretation of available information does not indicate the presence of major faulting, shearing, or significant folding in the interior portion of Dowe Flats.

5.5 Environmental Impacts

5.5.1 Surface Water

The mine project area does not contain any perennial streams. Only minor intermittent drainages are present. Storm runoff from disturbed land will be diverted and directed into the mined out areas such that no discharge to the St. Vrain Creek occurs (i.e. discharge resulting from runoff from disturbed land) or will be directed into a sediment pond. Any runoff from the

project area to drainages feeding St. Vrain Creek will meet applicable water quality standards.

Runoff from disturbed land will not be allowed to drain into any of the ditches present downgradient of the mining operation.

5.5.2 Air Quality

The applicant has installed an air quality/meteorological monitoring station on the property. Data have been collected pursuant to making an application for an air quality permit to the Colorado Air Pollution Control Commission (Colorado Department of Health). The operation will comply with all the terms, conditions, standards and monitoring requirements of that permit. Baseline monitoring data, communications with the State Health Department, air quality computer modeling and copies of the permit application will be forwarded to the County when they are available.

5.5.3 Land--Groundwater, Soils, Geology and Topography

A detailed discussion of groundwater is present in Exhibit G of the CMLRB application. The mining operation is not anticipated to impact groundwater quality or quantity. All applicable water quality standards will be complied with. Mining will not significantly impact drinking water wells in the area. Blasting vibrations will not damage domestic wells (see Blasting Technical Appendix). If necessary, a well permit will be obtained from the Colorado Division of Water Resources to comply with the requirements for ground water inflow into a mine and resulting evaporative losses.

Soils, geology and topography have been previously discussed.

5.5.4 Flora

Flora has been discussed in the Section on Vegetation.

5.5.5 Wildlife (Fauna)

Existing Conditions

Wildlife issues were originally described by Western Ecosystems, Inc. (1987). This report is attached as an appendix to this application.

Discussions with Boulder County, the Colorado Division of Wildlife (CDOW), U.S. Fish and Wildlife Service (USFWS), and the Colorado Mined Land Reclamation Division (CMLRD), indicate that as a result of Dowe Flats wildlife analyses, studies, and existing data, a wildlife data base exists that is adequate to meet local and state permitting requirements. Existing data includes CDOW WRIS maps and Latilong reports, the Rabbit Mountain Deer Study by Boulder County Parks and Open Space, wintering raptor and cliff nesting raptor studies by the Boulder County Nature Association, and 12 years of Longmont Christmas Bird Counts in the project vicinity. To complete this data base, additional burrowing owl and black-footed ferret surveys are scheduled to occur in summer 1993 and winter 1993/94, respectively.

Habitat

Study area characteristics, physiography, habitat and vegetation types present in the Dowe Flats area have been described by ESCO Associates (1987) and Western Ecosystems Inc. (1987). Significant habitats are shown in Map 6.

The marginal croplands and intensively utilized pasture on the permit area support below average wildlife richness and abundance values. However, some of these low quality habitats support large prairie dog towns which are ecologically important because of the prey base they represent, particularly for a large number of wintering raptors.

Relatively undisturbed habitats surrounding the project area (Rabbit Mountain, Indian Mountain, St. Vrain Creek Corridor) support rich wildlife communities. Daily and seasonal wildlife movements between these areas cross the project area and add to the importance of the site.

Wildlife Species

Two primary wildlife issues are associated with the project area: prairie dogs as raptor prey base and big game.

The prairie dog/raptor issue is clearly the most ecologically important wildlife issue associated with the project. Holistic Resource Management (1987) located 3,142 active prairie dog burrows on approximately 869 acres in 1987 and prairie dogs have expanded their distribution in recent years. This colony is important to a relatively large number of bald eagles (a state and federal endangered species) and ferruginous hawks (a federal candidate species) that winter in and around Dowe Flats. Other terrestrial and avian predators also exploit this prey base. Associated with this issue are the eagle and hawk roost/hunting perch sites in cottonwoods flanking the Supply and Highland irrigation ditches (refer to Map 6).

Big game, including mule deer, white-tailed deer, and elk, are seasonally present in and around the project area. Mule deer are the species of greatest concern, since they utilize Dowe Flats to a greater extent than the other big game species. Mule deer use Dowe Flats as an east-west movement corridor and, to a limited extent, as winter range.

Other sensitive wildlife-related issues associated with the project area include the riparian corridor and nongame values along St. Vrain Creek, potential destruction of raptor nest trees, and general loss of wildlife habitat.

Project Impacts

This section describes impacts to wildlife during and after mining and reclamation operations. Description of wildlife impacts considers the implementation of avoidance, minimization, mitigation and enhancement measures outlined in the reclamation plan. Discussions are detailed for those wildlife species and groups that are of special concern, and are less specific for other wildlife groups (refer to Exhibits E, F, & H in CMLRB application.)

Raptors

Although a wide variety of raptors seasonally hunt Dowe Flats, specific concerns focus on bald eagles and ferruginous hawks because of the permit area's importance to these endangered and candidate species. Mining impacts to raptors will be entirely indirect, as the result of reduced prey base availability. This reduction will be most significant during winter (November through March) when relatively large numbers of these birds focus on the permit area's large prairie dog towns. Therefore, the focus of the raptor issue is the availability of prairie dogs as prey base for raptors throughout the mining process.

Prairie dog towns in the project area were initially documented by Holistic Resource Management (1987). All acreages used in this section were updated in 1991 and will be updated in 1993, but are still approximate due to the dynamic nature of prairie dog distribution and a conservative estimate of the acreage of reclaimed areas that would be suitable for prairie dog habitat.

Prairie dog distribution in the valley is as widespread as it has ever been in recent years, due in part to reduced prairie dog control efforts by the present land managers. Prairie dogs have been controlled for decades in this valley by poisoning, plowing, shooting, and by other methods. As a result of the recent reduction in prairie dog control efforts, and due to the fact that Dowe Flats has avoided recent plague epizootics which periodically decimate local populations, prairie dogs are presently distributed throughout most of the valley.

Prairie dog distribution has recently expanded from four towns in the valley, and now totals at least 1,547 acres across the bottom of Dowe Flats. The largest town, running the length of the valley, totals at least 800 acres and is the only town that will be directly or indirectly affected by mining, reclamation, and road realignment. Over the 25 year life of mining activity, much of this town will be affected by mining, although no more than 12% at any given time.

As mining proceeds up the valley, reclamation will follow. Reclaimed landforms, specifically designed to accommodate prairie dogs and raptors, would be rapidly colonized as early successional vegetative communities provide ideal prairie dog habitat. A minimum of 30% of the affected acres would be reclaimed as suitable habitat for prairie dogs, lagomorph and deer habitats. Additional reclamation features (e.g. rimrocks) and reclaimed habitats would be

oriented toward enhancing lagomorph and deer habitats, additional prey species for the large raptors.

As mining begins and progresses, individuals in portions of the large prairie dog town would either be killed or would colonize adjacent unoccupied habitats. Raptors hunting these towns for prairie dogs and other resident prey species may decrease in numbers due to direct and indirect influences of mining activity, including reduced prey availability, avoidance of prey in habitat adjacent to mining and human activity, and the disturbance and temporary abandonment of roost sites, hunting perches, and hunting areas adjacent to haul roads and the relocated county road.

Post-reclamation prairie dog habitat in the valley would total approximately 75% of that now present. Additional suitable, but presently unoccupied adjacent habitats within the permit area, would also be available for colonization during or after the mining process, but are not included in these totals. The reclamation plan design spatially isolates post-mining prairie dog colonies to a greater degree than at present, thereby reducing the probability that a plague epizootic, which routinely decimates prairie dog towns along the Front Range, would wipe out the entire raptor prey base in Dowe Flats.

In addition to the perpetuation, specific reclamation, and isolation of large prairie dog towns, implementation of the raptor mitigation and enhancement measures discussed below are expected to more than compensate for any values lost during the mining process. These measures are summarized under Mitigation.

Big Game

Although mule deer, white-tailed deer, and elk are seasonally present in the vicinity of the permit area, big game concerns focus primarily on mule deer because of the relative importance of the permit area to this species. Elk winter range overlaps the western and northern portions of the permit area, but these areas of intensively utilized pasture and rotated agricultural fields are generally fallow in winter and are not used by elk during most years. White-tailed deer in the vicinity of the permit area are generally associated with the riparian habitats flanking the irrigation ditches to the south of the permit area. Accordingly, the basic big game mitigation assumption is that if the seasonal needs of mule deer can be met on and through the permit area,

those of elk and white-tails will also be protected.

The primary importance of the Dowe Flats permit area to mule deer is as a broad, east-west movement corridor that presently extends along the length of the valley. Important winter ranges on Rabbit Mountain are accessed by deer moving east across Dowe Flats in the fall. This pattern is reversed in spring as some deer migrate west to higher elevation summer ranges.

Winter range values of Dowe Flats are marginal because native vegetation is heavily utilized by cattle and prairie dogs, leaving little standing biomass during winter, and because agricultural crops are generally fallowed for the winter. Mule deer, which use portions of Dowe Flats during winter, take pressure off the heavily utilized Rabbit Mountain winter range. The intensity of utilization on Rabbit Mountain has been exacerbated in recent years by an increasing deer population and the CDOW's difficulty in managing this herd because the area and surrounding lands are closed to all hunting or to public hunting.

The proposed mining would have little adverse effect on the local deer herd. Although deer would be locally restricted from active pits and from areas awaiting reclamation, local and seasonal movements across Dowe Flats would continue. Broad corridors north and south of active mining and within the existing corridor would be more than adequate to accommodate continued movements.

The only obstacles that would restrict east-west deer movements from portions of their present corridor would be unreduced highwalls and ponds/lakes. However, neither of these features would adversely affect deer movements, because configuration of reclaimed landforms in these areas have been specifically designed to facilitate east-west movements. Valleys, buffered from the surrounding area by ridges, and vegetated with valuable big game foraging and cover species, should allow for leisurely, undisturbed residency as well as movements. The existing corridors also extend north and south of the permit area, so it would not be necessary to accommodate all future movements within the permit area. If fencing is constructed to restrict deer from reclaimed areas until after the vegetation can withstand big game grazing, east-west corridors would be provided between large fenced reclaimed areas to allow for continued movements.

Mining activity is not expected to significantly increase traffic volumes on County Road 47, and

no increase in big game highway mortalities is expected along the relocated County Road 47 or Highway 66. Haul truck activity between the mine and plant would be at 35 mph or less, speeds at which big game-vehicle collisions are infrequent. The movement corridor south of the proposed mine is sufficiently wide through unrestricted habitats to allow for continued movements. Disturbances to deer in this corridor will occur primarily from non-mine related activities and are not expected to force deer south onto Highway 66.

Threatened and Endangered Species

With the exception of the aforementioned effects and mitigation for bald eagles and ferruginous hawks, there is no evidence to suggest that any terrestrial state or federal threatened, endangered, or candidate species, including the whooping crane, sandhill crane, peregrine falcon, interior least tern, piping plover, white-faced ibis, western snowy plover, mountain plover, long-billed curlew, Preble's meadow jumping mouse, black-footed ferret, or swift fox would be adversely affected by the mining and reclamation option proposed for this project. Surveys for black-footed ferrets will be conducted in Dowe Flats during winter 1993/94 following USFWS (1989) protocol.

Other Wildlife Groups

Other wildlife groups, including small mammals, nongame birds, waterfowl and shorebirds, terrestrial predators, raptors, and fish, will experience minor impacts during mining, but will significantly benefit from the highly productive habitats that are created following mining. Although mitigation and enhancement measures would focus on the needs of raptors and big game, the diverse post-mining habitats would be exploited by other wildlife groups, greatly improving present trends of wildlife diversity and abundance.

Representatives of some wildlife groups now inhabiting the area will be displaced from localized portions of Dowe Flats for the life of the mining effort. Moreover, some species (e.g., badger, ring-necked pheasant, and vesper sparrow) may have a slightly smaller acreage of suitable habitat following mining and reclamation than now exists. However, none of the wildlife species now present would disappear and the greater diversity and abundance of wildlife supported by the post-mining habitats should outweigh the reduced abundance of species with affinities toward more common habitats of crop land and intensively utilized pasture.

MITIGATION

The mitigation discussion in this section refers to the Reclamation Plan which is found in Exhibits E and F of the CMLRB permit application.

Raptors

Proposed raptor habitat mitigation and enhancement measures include:

- Avoidance of the bald eagle roost/ hunting perches in the mature-decadent cottonwoods flanking the irrigation ditches east of the present county road.

- Establishment of a large cottonwood stand and numerous smaller stands that could eventually serve as a bald eagle roost. Although only a few cottonwoods would be removed from the haul road corridor crossing the ditches, it is expected that vehicular disturbance could displace eagles from a maximum zone within approximately 100 yards of each side of these roads (a 200 yard corridor). This distance is based on observations of perched, roosting, and hunting bald eagles adjacent to similar active mining operations at Southwestern's Lyons mine, approximately two miles south of Dowe Flats. It is expected that portions of the present roost would remain viable, although its capacity could be reduced. To enhance future roost availability in Dowe Flats, a replacement cottonwood stand would be established within the reclamation mosaic in an area visually (minimum 0.25 mile buffer zone) screened from potential disturbances outside the reclaimed landscaping.

- Installation of raptor hunting perches throughout the affected area. Initially, one to three telephone poles would be established atop reclaimed landforms to facilitate aerial predation. Poles would be established along the perimeter and within the interior of reclaimed areas overlooking prairie dog towns, reclaimed rimrock areas, and those habitats that are expected to be colonized by prairie dogs. Eventually, cottonwoods and ponderosa pines planted in shallow drainage basins atop and between landforms would naturally replace the function of the artificial poles. Bald eagles currently roost in and hunt from cottonwoods in an existing mine pit, south of the Southwestern plant.

- Creation of landforms and water features with diverse habitats supporting other (non

prairie dog) prey species (e.g., small rodents, lagomorphs, nongame birds, and waterfowl). The additional raptor hunting opportunities provided by the wildlife prey base in these habitats could compensate for reduced prairie dog availability. (refer to reclamation plan).

- Creation of rimrock habitat for lagomorphs, potential prey species for bald and golden eagles, ferruginous hawks, and other raptors.

- Creation of landforms which visually screen the relocated county road from raptor hunting, roosting, and nesting areas. This measure will make the prey base more available to hunting raptors.

- Retention of unreduced highwalls in appropriate areas to provide suitable sites for cliff-nesting raptors. No such cliff nesting habitat presently exists in Dowe Flats. There would be up to 2,500 feet of highwalls, 100 to 130 feet tall and 1,000 feet of highwalls, approximately 30 feet tall. Such cliffs, surrounded by proposed, complementary landforms/vegetation types and visually screened from most of the Dowe Flats valley bottom, could realistically be occupied by great horned owls, American kestrels, prairie falcons, golden eagles, ferruginous hawks, red-tailed hawks, barn owls, and common ravens. Depending on final highwall characteristics, "potholes" could be created (1 per 100 linear feet of cliff) to create additional nesting ledges. Other raptors, such as long-eared owls and sharp-shinned and Cooper's hawks, could also nest in adjacent habitats. These cliffs would also be used as nest sites for potential prey species (e.g., rock doves) and other wildlife ecologically important to the community (e.g., swallows). Refer to reclamation plan discussion and reclaimed landforms.

No burrowing owls have been observed in Dowe Flats, however a systematic replicated survey for burrowing owls will be conducted between May and August, 1993. If owls are located, appropriate mitigation measures that are acceptable to Boulder County and the proponent, will be cooperatively developed and implemented.

Big Game

After mining and reclamation, Dowe Flats would support much more valuable seasonal big game

habitats than are now present. Diverse landforms would support a variety of native vegetation types (e.g., grasslands, shrublands, herbaceous and shrubby wetlands, thickets and groves of trees) seasonally important to big game and other wildlife groups (see reclamation plan). Buffering landforms, cover, forage, and additional water sources will not only facilitate and enhance movements and winter range values, but also provide suitable fawning, transitional, and summer habitats. It is likely that the post-mine community would support a resident deer herd.

Threatened and Endangered Species

With the exception of the aforementioned effects and mitigation for bald eagles and ferruginous hawks, there is no evidence to suggest that any terrestrial state or federal threatened, endangered, or candidate species, including the whooping crane, sandhill crane, peregrine falcon, interior least tern, piping plover, white-faced ibis, western snowy plover, mountain plover, long-billed curlew, Preble's meadow jumping mouse, black-footed ferret, or swift fox would be adversely affected by the mining and reclamation plans proposed for this project. A black-footed ferret survey will be performed in the winter of 1993/1994 and the results will be forwarded to the County Parks and Open Space Department.

Other Wildlife Groups

Although mitigation and enhancement measures would focus on the needs of raptors and big game, post-mining conditions and opportunities would be exploited by other wildlife groups, greatly improving present levels of wildlife diversity and abundance.

In addition to previously mentioned mitigation measures, landforms (e.g., convoluted ridges, incised valleys, creeks, ponds, cliffs, islands, and gently rolling topography) will be created that support a diversity of vegetation types and a correspondingly rich wildlife community. Components within these landforms will include grasslands, shrublands, herbaceous and shrubby wetlands, thickets and groves of trees, nesting cliffs/ drop-offs for fisheries, spawning beds, waterfowl nesting/ roosting islands, and diving and dabbling duck foraging areas. The goal of reclamation within the former mine pits is to create an ecological community and setting similar to that of White Rocks along Boulder Creek, considered to be one of the most diverse and unique wildlife habitats in Boulder County.

5.5.6 Cultural Resources

Archaeology

Existing Conditions

The Dowe Flats region has received considerable attention from both professional and amateur archaeologists, beginning with Jack Moomaw (Burney 1989a). Archaeological resources were being recorded at Dowe Flats as early as 1949. Subsequent inventories were undertaken by the Lyons Chapter of the Colorado Archaeological Society during the 1970s and early 1980s. The documentation by Nykamp (1984a and 1984b) describes the Rabbit Mountain survey conducted by the Lyons Chapter of the Colorado Archaeological Society.

Thirty-two prehistoric and historic sites and isolated finds have been previously recorded in or near the Dowe Flats project area, primarily on Rabbit Mountain. Several sites have also been recorded west of the project area on Indian Mountain, including a presumed archaeoastronomy site, tipi rings, and other stone features.

The only formal excavations conducted near the study area were completed by Cassells and Farrington (1986) at site 5BL876, the Indian Mountain site. This multicomponent site consists of at least 10 stone circles. Stone flakes and a ceramic sherd were recovered at this site.

Although a wealth of archaeological sites have been recorded near the project area, definitive analysis and clear determinations regarding the significance of these sites is generally lacking. Detailed site descriptions with surveyed boundaries are often absent. Only the one site on Indian Mountain has undergone subsurface testing.

Virtually no systematic, detailed archaeological inventory or comprehensive report on the prehistory has ever been completed for the Dowe Flats region as a geographic unit. Without having comprehensive information treating the whole of Indian Mountain, Dowe Flats, and Rabbit Mountain, the cumulative significance of the local prehistoric resources cannot be determined. Burney (1989a, 1989b), as a part of the initial Dowe Flats project survey, has taken the first step to comprehensively report on these resources, and recommended specific actions to continue this process. Additional archaeological and historical studies are being undertaken

on selected portions of the Rabbit Mountain Open Space (Gleichmen 1992). In addition, upcoming consultations with various Indian tribes, regarding the proposal, may provide significant information on the past use of Dowe Flats by Indian peoples.

Based on these recent surveys and research of archaeological site files and literature, it is well documented that Dowe Flats had been intensively occupied for thousands of years by Indian peoples. Archaeological studies conducted by Burney and Associates, Inc. are attached as an appendix to this application.

State of Archaeological Resources

In studying the archaeological history of Dowe Flats, one is impressed with the number of prehistoric artifacts that have been collected and removed from the area since historic settlement began. Despite this irretrievable loss, archaeological resources still exist.

Dowe Flats has been under cultivation for many years and agricultural practices have impacted the archaeological resources, particularly the integrity of intra-site configurations. As noted by Moomaw (1960):

> "When I homesteaded on Rabbit Mountain in 1915, there were seven tipi rings near the cabin. The land to the west of the homestead was plowed up, and through the years wind-blown soil has completely covered these seven rings. I also destroyed dozens of rings while clearing fields for farming. This same process has gone on in the vast areas now being farmed in this region..."

Meier (1987) notes the impacts of the Town of Noland on archaeological resources in the Indian Mountain area from historical documentation:

> "Sundays at Noland were generally lazy days. Many of the men climbed the high ridge to the east [Indian Mountain] and searched for old Indian graves."
>
> "There was, at one time, a lot of Indian writing on the cliffs at the top of Indian Ridge...but most of it has been chipped off and hauled away."

"The children's out-of-school hours were spent in search of adventure and Indian relics."

"One of the men managed to find a bit of adventure in one of the Indian hideouts when he had a narrow escape from a wild cat while hunting Indian beads."

Given the intense activity in the Noland and Lyons area during the height of the quarrying era, it is remarkable that any prehistoric cultural materials remain on Indian Mountain, and that the rings and structures on both Indian Mountain and Rabbit Mountain were not severely disturbed.

Through the cumulative effects of collecting, defacing, development, plowing, and other disturbances, the prehistoric resources of north-central Boulder have been diminished. The state of archaeological resources is regrettable both in terms of the loss of significant amounts of material, and the integrity of what remains. As archaeological resources become increasingly diminished, the long term conservation of the remaining resources becomes even more significant.

Project Surveys and Documented Resources

Table 5.5.6-1 lists twelve archaeological projects completed between April 18, 1989, and February 3, 1991 in conjunction with this permit application. In all, approximately 1550 acres have undergone intensive Class III pedestrian surveys. In addition to the field surveys, documentary research was conducted at the State Historical Preservation Office, Lyons Redstone Museum, Boulder Historical Society and Museum, and the University of Colorado Museum. Consultation regarding the American Indian resources known and yet to be discovered for Dowe Flats is being pursued with the Northern Ute, Ute Mountain Ute, Southern Ute, Northern Arapaho and Cheyenne, Southern Araphaho and Cheyenne, the Eastern Shoshone, and Commanche tribes.

As a result of the field surveys, two sites and 31 isolated finds have been recorded. Sites include 5BL2431 and 5BL3129. Site 5BL2431 underwent limited subsurface testing in 1990 and has been identified as a deeply buried campsite dating as early as 660 B.C. Project archaeologists have determined that the site is eligible for the National Register of Historic Places (Grant 1990). Site 5BL3129 was identified in 1990 and contains tipi rings. This site has not undergone detailed investigations.

Isolated finds include ground and chipped stone artifacts. These 31 isolated finds recorded throughout the project area demonstrate the disarticulated character of the once intact prehistoric remains present on the ground surface.

The mine plan avoids direct impacts of all known and recorded archaeological sites. Site 5BL2431 is approximately 850 feet from the nearest mine and reclamation activity. Site 5BL3129 is more than 800 feet from mining and reclamation activities and adjacent to reclaimed overburden deposits. Of primary concern then, is the potential for the presence of subsurface sites throughout the mine and reclamation area. Since these sites can not be detected until earth moving activities begin, it is recommended that a specific mitigation process, as described below, be agreed to prior to start up of project activities. Monitoring of ground disturbing activities of soils potentially containing archaeological deposits is recommended as part of this process.

Mitigation

Because the majority of land in the Dowe Flats project area is owned by private parties or the Boulder County Parks and Open Space Department, it is assumed that Boulder County will be the lead agency in the management and disposition of prehistoric and historic cultural resources. There are no state or federal lands in the project area. The project does not require a federal "undertaking" with respect to Section 106 of the National Historic Preservation Act. However, the applicant has agreed to use this Federal regulation as a model in the mitigation and management process.

It is anticipated that Boulder County will work closely with the Colorado State Historic Preservation Office (SHPO) regarding the mitigation of adverse impacts. Mitigation or data recovery can include several measures, from avoidance to complete excavation.

There is a significant potential that archaeological materials may be exposed during mining activities, particularly in areas having adequate soil depth to support undisturbed cultural materials below the ubiquitous plow zone. A contingency plan to address this situation will be included in the Cultural Resources Management Plan. In addition, it is possible to adjust the boundaries of reclaimed landscape features without constraining the overall reclamation plan.

A formal process is proposed in order to identify and implement specific management and mitigation measures as the need arises, as displayed in Table 5.5.6-2.

Historic Resources

Project Area Surveys

Survey methodology followed two steps. The first step identified research sources and types of information available to support the historic context and property evaluations. Sources included a file search at the State Historical Preservation Office. Michael Burney's report (1989a) of his archaeological survey of Dowe Flats proved of great utility. Bibliographic searches were undertaken at local repositories such as Colorado Historical Society, Denver Public Library, University of Colorado Library, water companies, and contacts with county offices (recorder, assessor, county agricultural extension agent). The documentary information collected from these sources was evaluated and developed into the historical context of the project area. Such work met all requirements for a historic context as specified by the Colorado Survey Manual and National Register of Historic Places Bulletin 16.

The second step of the methodology was site specific field research, to fulfill the requirements of a Class II level survey under Colorado guidelines and applicable Secretary of the Interior standards. From this research, data for documentation and management recommendations for the properties was generated. The information was used during form preparation, final context writing, survey report writing, and recommendations of site eligibility for the National Register of Historic Places. The cultural history report is attached as an appendix to this application.

Historic Context

The history of the lands that comprise the Dowe Flats area is dominated by the evolution of a high plains, rural agricultural lifestyle. Other factors, particularly mining and quarrying, influenced the area's history. The pertinent historic themes for Dowe Flats include:

1.	exploration and fur trade	1700-1845
2.	Colorado gold rush and early settlement	1858-1870
3.	early agricultural and ranching development	1870-1895

4.	quarrying and urban growth and development	1870-1900
5.	ranching and farming	post 1900
6.	the Great Depression and World War II	1929-1945

Project Survey Results

Class I and Class II Dowe Flats historic surveys (Western Historical Studies, Inc., 1990) identified and recorded six buildings, structures, and features. Of the six resources of sufficient age (50 years or older) to be considered as possibly significant historic resources, four of the resources were irrigation ditches (Supply, Highland, Rough and Ready, and Palmerton); one was an equipment shed and the other was a thresher or corn sheller and an associated jumble of used fencing. The four ditches are recommended as eligible for inclusion in the National Register of Historic Places (NRHP) with appropriation dates of 1862. All of the ditches have a high degree of integrity, convey the feeling of irrigation supply ditches and represent a major theme of northern Boulder County agricultural development. Near the study area, along one of the ditches, is a ditch walker's shanty.

The Class I study identified two other resources previously determined eligible for the NRHP. These were the Montgomery School (5BL242) and the Gerstenkorn Homestead (5BL241) located on nearby Colorado 66.

Project Impacts

Primary Impacts

Construction and use of the Mine Haul Road may have potential minor impacts on the four irrigation ditches. Impacts include placement of new bridges and culverts. A ditch walker's shanty associated with the Rough and Ready Ditch is within one of the proposed transportation corridors, and may be impacted depending upon the final haul road alignment. There are no significant resources in the area disturbed by mining and reclamation.

Secondary Impacts

The Montgomery School, and other presently unrecorded resources along Highway 66, are located in areas that are not expected to be subject to secondary impacts from construction and mining activity. Blasting vibrations from the Hi-Cal pit on Limestone Ridge are not expected to have significant impact on the Montgomery School. Vibrations from blasting and haul trucks have been described in the Blasting Technical Appendix and the project historian is evaluating this information. The historian's report will be forwarded to the County.

Mitigation

Because the four ditches are similar in function and character, a mitigation plan for one has been developed that can be applied to all. Should these resources become subject to adverse impact, a formal request should be made to the State Historic Preservation Office (SHPO) for a concurrence of the eligibility recommendations and an agreement to a mitigation plan for the resources. If there is federal participation, a consultation process with the Advisory Council on Historic Preservation must be initiated by the federal agency to gain council comment on the proposed mitigation efforts. For those resources recorded in the historic resources survey and determined to be eligible, the most appropriate mitigation plan would be to complete a permanent record of the sites. This record might be limited to medium or large format archival photography and more detailed histories of the properties. Above that level, a Historic American Engineering Record (HAER) study of each resource could be performed. A HAER study typically includes 4" x 5" archival photography, preparation of detailed drawings and writing of a narrative history of the property. Such studies have been used to successfully

mitigate other irrigation ditches in Boulder County.

For the Montgomery School and other presently unrecorded resources along Highway 66, no mitigation plans are offered at this time. These resources are sufficiently removed from the direct impact areas so that no plans, other than avoidance, need be addressed at this time. Any of these resources could be incorporated into the Cultural Resources Management Plan process for long term monitoring.

5.6 Adequate Water Supply

Operations Center

The operations center for the Dowe Flats project will be housed in the existing structures on the Harroun farm. The existing water supply on the Harroun farm will have to be supplemented with the applicants' existing water rights in the area to supply potable and production water (e.g. water for dust control).

5.7 Adequate Sewerage System

The operations center for the Dowe Flats project will be housed in the existing structures on the Harroun farm. The existing sewerage system may have to be enlarged to accommodate increased use by mine employees.

5.8 Utility Relocation Corridor

As discussed in Section 5.10 below, County Road 47 must be vacated. The road relocation will affect buried telephone cable belonging to U. S. West and the mining and reclamation will affect power lines owned by Poudre Valley Rural Electric Association (PVREA). Negotiations are in progress between the applicant and U. S. West regarding relocation of the telephone cable.

PVREA is suggesting the applicant include a utility relocation corridor for the power lines as part of Boulder County Special Use permit approval. The proposed new 200' wide corridor, agreed upon between PVREA and the applicant, is shown on Map 1 of this permit application. It will cross the southern portion of the mine permit in an area not disturbed by mining.

Power poles will be installed so as to avoid the mapped wetlands in the vicinity of the new utility corridor. The power line corridor was specifically designed to minimize impacts to cottonwoods and shrub understory along irrigation ditches.

5.9 Government Services

Government services are not required for this project.

5.10 Transportation Impacts

State Highway 66

Access and crossing of State Highway 66 will be subject to the State Highway Access Code. An access permit will be applied for and all required terms, conditions and mitigations will be implemented. Plans for the access and crossing, as well as any required mitigation, will be developed by the Colorado Department of Highways in conjunction with the Boulder County Public Works Department.

County Road 47

For the purposes of this project the existing County Road 47 (North 53rd Street and North 55th Street) must be vacated. The vacation is requested because North 53rd Street will be impacted by the Hi-Cal quarry, and North 55th Street will be impacted by the Third Ridge quarry. In addition, it is not considered appropriate or safe for a public road to run through the middle of a quarry operation. In addition, the County Road is within the reservoir influence zone, and at such time as a reservoir is constructed, it would be necessary to relocate the road.

The vacation of County Road 47 is proposed to begin approximately one half mile north of State Highway 66 and relocated into the West Valley of Dowe Flats. The one half mile of the road to remain north of State Highway 66 will serve as local private access and mine access (Map 1).

The vacation is proposed to end where the St. Vrain Supply Canal crosses the existing North 55th Street. The section of road north of this point may serve as access to the Rabbit Mountain

Open Space and the St. Vrain Supply Canal.

That section of North 53rd Street south of the St. Vrain Supply Canal and south to where the boundary of the Rabbit Mountain Open Space turns east may remain as restricted local access for the applicant and Rabbit Mountain Open Space properties, or removed entirely.

Numerous alternatives were evaluated in consideration of the new alignment of County Road 47. Alternative alignments evaluated in addition to the preferred alternative include:

1. North 61st St. north along the western slopes of Rabbit Mountain to the north end of Dowe Flats

2. Using the preferred alternative with a new access point on Colorado Highway 66 between North 53rd St. and Highland Drive

3. Highland Drive north through the western edge of the West Valley

4. Highland Drive and the St. Vrain Supply Canal maintenance road.

The preferred alignment presented in this proposal is based upon and best satisfies the following criteria:

1. Avoidance and minimization of impacts to environmental constraint areas including wetlands, archaeological sites, wildlife habitat, and open space

2. Spatially separating the mine and reservoir operations in the East Valley, and the Mine Haul Road in the West Valley and on Limestone Ridge

3. The need to maintain a safe access from State Highway 66

4. Preference for the shortest, most direct route

5. Avoidance of costly engineering and maintenance (tight corners, steep grades, large cuts & fills, deep shade)

6. Preference to maintain the right-of-way in Southdown's ownership

7. Minimization of scenic impacts in the Dowe Flats area

8. Minimizing neighborhood impacts from the realigned County Road 47.

9. Recognition that Boulder County Department of Transportation and Colorado Department of Transportation will review alternatives and determine the final alignment.

Transportation Analysis

A detailed traffic impact analysis was performed for various hauling alternatives. Public road hauling and private road hauling were considered, as well as 25 ton and 85 ton haul truck scenarios. The preferred alternative is a private haul road west of the existing plant entrance road (County Road 49) using 85 ton haul trucks. Southdown prefers the larger trucks because it involves fewer crossings and therefore less potential interference with Highway 66 traffic. The larger trucks are also more economical for plant operations. Southdown also desires to have the crossing signalized, and activated only at the times of actual crossings.

Traffic related to plant capacity, employees, suppliers, etc. will not increase because the Dowe Flats Mine will not increase plant output, it will extend the life of the plant. Employee traffic to the proposed mine will only be shifted from other Southdown locations. Also, approval of full production at Dowe Flats will eventually eliminate 12,000 truck trips per year which currently haul limestone for 11 miles on county roads and on State Highway 66, from the Larimer County quarry.

The traffic analysis indicates that even during the peak season there will be sufficient gaps in the flow of existing traffic to allow trucks to safely cross Highway 66 at grade. The crossing does not meet the requirements for a traffic light although Southdown wants to install one. To enhance safety and operations, the following measures could be considered:

- The haul road should be restricted to haul trucks only, perhaps using signs and gates.
- Advance warning signs and flashing beacons should be installed on State Highway 66 to advise motorists of trucks entering.
- The intersection should be illuminated even though no night hauling is proposed (darkness comes early in the winter months).

- The 20-year traffic projection indicates gaps may not be adequate 20 years from now. If this happens, hauling could be limited during peak traffic hours or the intersection could be signalized.

The transportation analysis is in the Technical Appendix.

6.0 REFERRAL

A comprehensive permit referral list was developed in cooperation with the Boulder County Planning Department. The referral list includes a total of 75 entities that will be informed in detail on the permit applications as follows:

Regulatory and Planning Agencies	
Municipal & County	7
State & Federal	10
Neighborhoods	10
Utilities	3
Environmental, Planning and Community Organizations	13
Indian Tribes	9
Media	6
Mine Industry Organizations	9
Water Providers	8

The referral list is updated and modified on a monthly basis. The referral list has been developed into a permit document matrix and attached to this permit application.

6.1 Community Information Program

To supplement the County permit referral process, the applicant developed a Community Information Program (CIP) to provide for direct communication with the interested parties on the current referral list, plus other entities or individuals that become involved in the permit process.

There are five structured phases to the CIP as follows.

Phase I

Introductory Meetings (prior to or at time of submittal of permit application)

These are 20-30 minute meetings consisting of a 10-minute presentation and 10-20 minutes of questions and answers. Content focuses on: 1) current plant operations, 2) introducing the Project Management and Coordinating team, 3) describing the Dowe Flats Project in broad terms. At this time, the applicant requests that each group appoint a Permit contact person, and that each group appoint a committee to review the forthcoming permit applications.

Phase II

Detailed Presentations (early in the referral period)

Approximately two hours or longer meetings where the project is described in detail. Presentations will be customized to the obvious concerns of each agency or entity. The purpose of Phase II meetings is to ensure that individual groups have the level of information necessary for them to make reasonable evaluations of the permit application.

Phase III

Meetings to Consider Interest Group Concerns (late in the referral period)

These meetings are designed to hear the concerns of each agency or citizen group. Ideally, each group will have heard the Phase II detailed presentations, their committees will have read and

evaluated the permit applications and appropriate technical appendices, and will be prepared to express specific concerns regarding the project. The goal of Phase III is to ensure that each agency or citizen group concerns have been heard by the Project applicant and consultants.

Phase IV

County Staff Review (after the referral period, and prior to final staff recommendation)

After the close of the referral period, staff will evaluate referral comments as well as incorporate their own issues into a final staff recommendation. Ideally, each of the agencies and citizen groups that the Project team approached during the referral period will have expressed their concerns by this time and additional mitigation plans may be offered at this time. The goal will be to address the address the reasonable concerns expressed during the referral period into the permit application.

Phase V

Final Meetings (after final staff recommendation, prior to public hearings)

These are follow up meetings with individual groups to discuss how the concerns expressed during the referral period have been addressed and incorporated into the permit application.

Some entities on the Community Information Program List may only request a Phase I meeting. It is anticipated that other entities, such as nearby property owners and environmental organizations may need several additional meetings, beyond what is described herein, in order to address their concerns. The CIP will be customized for each entity as necessary.

Interspersed during and after Phase II may be field trips to the Dowe Flats area, as requested by the citizen groups.

Prior to permit application submittal on June 1, the applicant had:

- completed Phase I CIP meetings with 38 of the 75 entities on the referral list,

- contacted an additional 23 entities and requested Phase I CIP meetings,
- mailed letters on May 3 to all 250 adjacent property owners and other neighbors,
- on May 14 provided a press release to six area newspapers.

7.0 CONDITIONS AND STANDARDS FOR APPROVAL

7.1 Minimum Zoning Requirements Compliance

The project area is zoned Agriculture. The project is consistent with special uses permitted in the Agriculture Zone District as demonstrated in this permit application.

7.2 Harmony With Character of Neighborhood/Compatible With Surrounding Area

Description of the Neighborhood

The immediate neighborhood, generally a one mile zone reaching beyond the project boundary, is rural in character. Residential development is thinly scattered throughout the area, with only three small subdivisions within or near the one mile radius. The densest development is along the north side of Highway 66, approximately one-half to one mile south of the project boundary.

Dowe Flats is unique in Boulder County in that the valley bottom north of the Supply Ditch has no permanent residences. There is one residence at the north end of the valley as it rises up into the Dakota Hogback. Thus Dowe Flats presents an open, undeveloped landscape.

The neighborhood may be split into four geographic units based upon the natural setting of Dowe Flats. The properties described below are not under control of the applicant unless specified.

North

This end of Dowe Flats forms the head of the valley, turning steeply upward into the Dakota Hogback. The Larimer County border is approximately one mile north of the north end of Dowe Flats.

The Indian Gap Subdivision contains ten lots and is three-quarters to one mile northwest of the project area.

In addition to the Indian Gap Subdivision there are four properties north of the project area, to the Larimer County border.

The Indian Gap neighborhood mailing list maintained by the applicant has 42 property owners, indicating that a number of residences in Larimer County are served by County Road 47.

West

Indian Mountain forms the western boundary of Dowe Flats, rising 300 to 1000 feet above the valley bottom. South of the Indian Gap subdivision, there are 3 undeveloped properties on Indian Mountain. One is owned by the United States government and is considered surplus federal property. The 283 acre Card parcel is owned by the applicant.

East

East of the project boundary. Rabbit Mountain, rising 300 to 600 feet above the valley, forms the eastern boundary of Dowe Flats. There are 4 properties in this area. With approximately 1,300 acres, the largest neighboring land owner, in terms of total acreage, is Boulder County with the Rabbit Mountain Open Space.

Peper Ridge Farms, a 3 lot subdivision, is one and one-quarter miles to the east. This subdivision is completely screened from the project by the southern portion of Rabbit Mountain.

South

The south end of Dowe Flats is the open end of the valley, bordered by St. Vrain Creek. There are 24 properties bounded by the project boundary on the north, Colorado Highway 66 on the south, Highland Drive on the west, and North 61st Street on the east. The 5 properties to the west of the Limestone Ridge are screened from the quarries by Limestone Ridge. The 19 properties east of Limestone Ridge will be screened from the mine quarries by the gently rolling topography of the valley bottom and the construction of reclaimed landscapes. All four irrigation ditches lie between 8 of these properties and the project area.

South of Highway 66, the Southwestern's Lyons Plant is the major landholding. Along St. Vrain Creek are a few widely scattered residences and a number of active and reclaimed gravel mines.

The McCall Lake Subdivision with 8 developed lots is one mile to the southeast of the project area.

7.3 Accordance with Boulder County Comprehensive Plan

Applicable Goals and Policies for the Dowe Flats Project

Policy

Diverse, compatible, and functional land use patterns should be established and, when necessary, revised to prevent urban and rural decay.

Commentary

The project maintains mining, agricultural, and wildlife land uses in north-central Boulder County. Existing industrial facilities at the cement plant are maintained.

Policy

Within Community Service Areas a suitable balance between employment opportunities and available housing, in light of the labor force and other demographic characteristics of the community, should be established or maintained.

Commentary

The project is within the Town of Lyons Potential Service Area. The effect of this permit application is to maintain existing jobs at the Southwestern Portland Cement Company plant. The Lyons Comprehensive Plan Planning Policies direct that the town should maintain its small-town atmosphere while supporting a diversity of employment activities. There are no impacts on housing supply. The Lyons Comprehensive Plan anticipates mining in Dowe Flats.

Environmental Management Policy

Unique or distinctive natural features and systems and cultural features and sites should be conserved and preserved in recognition of the irreplaceable character of such resources and their importance to the quality of life. Other resources should be managed in a manner which is consistent with sound conservation practices, while enhancing compatibility between natural and man-made characteristics.

Commentary

The project preserves the function of existing features; no irreplaceable features are impacted. The reclamation plan is designed to mimic the topographic relief and wildlife habitat values of Rabbit Mountain. The agricultural and wildlife land uses developed through the reclamation plan are compatible with neighboring open space. Mining land use is compatible with cultural history and land use of Town of Lyons.

Policy

Air, water, and noise pollution; inappropriate development in natural hazard areas; and overall environmental degradation should be reduced as much as possible or eliminated in order to prevent potential harm to life, health and property.

Commentary

This project complies with state regulations governing air, water, and noise pollution. The project is designed with adequate buffers to protect the safety and health of the public.

Policy

Critical wildlife habitats should be conserved and preserved in order to avoid the depletion of wildlife and to perpetuate and encourage a diversity of species in the County.

Commentary

The reclamation plan greatly enhances the existing wildlife habitat. Functional movement and cross-valley migration corridors are developed for deer and elk. Adjacent to this project, on Rabbit Mountain, is critical habitat for white-tailed deer. The reclamation plan addresses this issue by maintaining big game movement corridors through Dowe Flats south to St. Vrain Creek and west to Indian Mountain. The majority of the valley-wide prairie dog habitat is not impacted, thus preserving and enhancing prey base for raptors.

Policy

Critical plant associations and rare plant sites should be conserved and preserved to encourage a diversity of plant types within the County and to perpetuate the species.

Commentary

No rare plants exist in the project area. Limestone Ridge was thoroughly searched for rare plants, particularly Bell's Twinpod, without positive results. The reclamation plan

accommodates future plantings of species that prefer shale and limestone outcrops. The reclamation plan conforms with USDA Soil Conservation Service recommendations for existing soil types. Vegetational diversity is greatly enhanced with the development of the reclamation plan.

Policy

Wetlands which are important to maintaining the overall balance of ecological systems should be conserved.

Commentary

Wetlands impacts are less than one acre total. The reclamation plan creates approximately 44 acres of open water and 37 acres of wetlands.

Policy

Unique or critical environmental resources shall be conserved and preserved in a manner which assures their protection from adverse impacts, with the private sector, non-county agencies and other governmental jurisdictions being encouraged to participate.

Commentary

The reclamation plan is implemented through the private sector without the use of government funds. The applicant has participated in private sector land conservation on nearby lands in the Foothills Highway (US 36) Scenic Corridor where 175 acres have been removed from mining, with additional restrictions applying to development and protection of wetlands.

Policy

Productive agricultural land is a limited resource of both environmental and economic value and should be conserved and preserved.

Commentary

The vast majority of the project area is constrained by soil types suited only for range and wildlife habitat land uses. The 160 acre Harroun parcel is the only significant agricultural parcel (lands of state-wide importance). Most of this parcel is not impacted by the project, and will remain in agricultural use.

Parks and Open Space Policy

Provision should be made for open space to meet human needs throughout the County in order to protect and enhance the quality of life and enjoyment of the environment.

Commentary

The mine plan buffers and setbacks provide open space for the 25 year life of the mine operation. The reclamation plan is designed to mimic the topographic relief and wildlife habitat values of the neighboring Rabbit Mountain Open Space.

Policy

Open space should be promoted as an urban-shaping method and as a means of protecting from development those areas which have significant environmental, scenic or cultural value.

Commentary

The reclamation plan will serve as a buffer to the Rabbit Mountain Open Space. The project area is functionally integrated with Rabbit Mountain to provide wildlife movement corridors across Dowe Flats.

Policy

A County-wide trail system shall be promoted to serve transportation and recreation purposes.

Commentary

The proposal has no conflicts with the St. Vrain Supply Canal and the St. Vrain Creek trail corridors.

Residential Goals

[none applicable]

Community Facilities Policy

Development should not place any undue burdens on any existing community.

Commentary

Minimal community services are required. The proposal maintains existing jobs and industrial facilities.

Economic Conditions Policy

A balanced, diversified economy should be encouraged in order to maintain and enhance the quality of life of Boulder County citizens by assuring desirable local employment opportunities and strengthening and stabilizing the tax base.

Commentary

The Southwestern Portland Cement Company's Lyons plant provides 83 jobs with an annual payroll of $3.5 million. Southwestern spends an average of $10.5 million per year for supplies and materials. The mine permit provides for the continued operation of the existing plant, approximating historical levels of output. The plant pays a $850,000 per year in property, personal, and equipment taxes to Boulder County agencies, but utilizes few community services.

Policy

Commercial or industrial development should provide an attractive, safe and healthy environment which does not have an adverse impact upon the surrounding area.

Commentary

The existing cement plant has been a neighborhood feature for 24 years. The mine proposal will phase out 6000 individual 25-ton truck trips per year on an 11 mile haul route on county roads and Colorado Highway 66. The Mine Haul Road has been designed to minimize conflicts on County Road 47 and Colorado Highway 66. All applicable state and federal permits will be secured.

Policy

Rehabilitation of existing commercial and industrial facilities should be promoted where feasible. A climate for industry which is consistent with community growth desires and the ability of the community to absorb growth should be encouraged.

Commentary

The proposed land use is recognized in the Lyons Comprehensive Plan. The proposal maintains existing jobs and industrial facilities and is not expected to create conflicts with growth policies.

Transportation Policy

An efficient, coordinated, multi-mode transportation system, which will provide adequate and safe movement of people and goods, should be promoted while avoiding undesirable environmental impacts.

Commentary

The proposal will eventually remove existing mine traffic from N 83rd and 75th Streets, Colorado Highway 66, and Hygiene Road. In addition to vehicular traffic, N 83rd and 75th Streets and Colorado Highway 66 are designated bike routes and receive extensive use by bikers.

Policy

The aesthetic quality of the County should be given full consideration when locating, designing or upgrading any part of the transportation system, in order to preserve and enhance public enjoyment of such facilities as mountain roads, bridges, scenic overlooks and landscape plantings.

Commentary

County Road 47 will be designed in accordance with the specifications of the Boulder County Department of Transportation.

Public Involvement Policy

The County shall encourage citizen public participation in the making of decisions by public and quasi-public bodies which significantly affect citizens.

Commentary

In addition to the required public hearings and standard referral process, the applicant has proceeded with a Community Information Program that is unprecedented in scope.

Governmental Relations Policy

The County shall encourage and promote coordination and cooperation between Federal, State and Local Governmental entities charged with making decisions which significantly affect land used in Boulder County.

Commentary

All appropriate agencies are involved in the project; including the US Fish and Wildlife Service, US Army Corps of Engineers and Colorado Division of Wildlife.

7.4 No Overintensive Use of Land

The proposed use is not considered over-intensive.

7.5 Capital Improvement Program Impacts

No county funds are required for this project.

7.6 Governmental Services Impacts

No governmental services are required for the mining operation.

7.7 Traffic Congestion

Refer to discussion in Section 5.10 and the transportation analysis in the Technical Appendix.

7.8 Noise and Blasting

7.8.1 Noise

A detailed noise analysis was performed at Dowe Flats to characterize the ambient noise conditions and to predict the noise impacts from the proposed mining and hauling activities. Noise impact mitigation techniques are also proposed.

The applicable nose standard is State of Colorado noise law and it has been determined that these standards can be met (the stringent residential standards) at the property line except when reclamation work is occurring near that boundary. Slight exceedances of off-site noise standards will be experienced at these times during reclamation. The sculpting of reclaimed

landforms could be setback from the property line to prevent these noise standard exceedances but this could compromise the integrity and goals of the overall reclamation plan.

Noise emissions during mining of all four ridges, for the entire life of the mine, will not exceed noise standards at the property line. Haul truck noise emissions for the various hauling alternatives cause slight exceedances of noise standards at some off-site properties. These properties are, however, located along Highway 66 and the existing traffic noise will mask the haul truck noise to an extent. Mitigation for haul truck noise includes the installation of noise control kits, speed control, and prohibition of Jacobs Brakes. Haul trucks meet EPA noise standards for public highways. Additional noise mitigation includes use of topographic barriers (i.e. reclamation hills), noise control kits on mining equipment, and movable noise shields around rock drills.

The noise analysis is in the Technical Appendix.

7.8.2 Blasting Impacts

An evaluation was made of potential impacts of blasting operations at the Dowe Flats project on the surrounding area. The complete blasting analysis and evaluation is presented in an engineering report presented in the Blasting Technical Appendix.

Outside the mine permit area, the primary blasting impacts are the ground vibrations and air overpressure resulting from the blasting. To assess the magnitude of these affects, an analysis was completed of monitoring data developed at the Lyons Quarry which currently provides materials to the Southwest Portland Cement Plant. In that the blasting operations currently conducted at the existing quarry is in the same geologic setting at Dowe Flats and the proposed blasting operation will be essentially the same as at Lyons Quarry, the existing operation provides a very good representation of Dowe Flats blasting impacts.

Data developed at the Lyons Quarry was analyzed to develop predictive relationships between ground vibration and air overpressure levels as a function of distance from the blast and the amount of explosive used. With these relationships, estimations of the maximum level of air overpressure and ground vibration measured as peak particle velocity were made at various distances from the limits of the mine area. These estimates are considered conservative in that:

1. An upper bound limit of all measured blasting data was used in determining the predictive relationships used.

2. Blasting was considered to occur at the nearest point within the mine excavation limits.

3. The upper bound amounts of explosive weights per blast that has been historically used at Lyons Quarry were considered.

4. A more restrictive method of calculating the resulting peak ground motion was used than is required by current U.S. Office of Surface Mining Reclamation and Enforcement (OSRME) regulations.

With the degree of conservatism incorporated into the analysis, the resulting prediction of ground vibration and air overpressure are considered to be upper limit values (i.e. worst case). A significant number of blasts would induce offsite impacts below the projected levels.
Comparing the projected maximum values to standards established by OSRME, the maximum offsite effects are within the established limits. Ground vibration and air overpressure levels resulting from the proposed blasting can be maintained below levels that would induce building damage off the mine permit boundary. Because of the relative sensitivity of humans to the blasting induced impacts, there will be some level of human perception of the blasting away from the mine boundary, the comparative level depending on location and individual sensitivity.

7.9 Adequate Landscape, Buffer and Setback, Screening

Visual issues will be discussed under this section. An in depth visual analysis of the mine and reclamation project is described in the Visual Analysis technical appendix. The inherent visual mitigation features in the reclamation plan are presented in detail in Exhibits E and F of the CMLRB reclamation permit. The following discussion provides a summary of these reports as they relate to this section.

7.9.1 Landscape and Screening

The reclaimed landscape outside the active mine pits are specifically designed to screen mine

operations from foreground views for approximately one half mile to the east, west and north, and will screen views to the south into the distant background (beyond one half mile) due to the flat relief in this direction. Screening is specifically provided for:

- County Road 47 from Colorado Highway 66 north to a point one quarter mile south of Dowe Pass,
- Colorado Highway 66 east of the County Road 47, including residences in this area,
- that portion of County Road 47 that will connect to the Rabbit Mountain Open Space,
- the parking lot and adjacent low lying areas of the Rabbit Mountain Open Space.

The phased development of the reclaimed landscape outside of the pit quarries will be linked to screening the active in-pit operations. This plan provides for screening of the majority of mine operations, however there will be occasional temporary visibility of mine operations at:

- the peak of Limestone Ridge, ahead of the advancing out of pit reclaimed landscapes,
- limited portions of the in-pit haul road on Limestone Ridge,
- most of the out of pit portion of the haul road.

7.9.2 Buffer and Setback

Through acquiring numerous properties in the Dowe Flats area since 1984, the applicant has been able to provide for extensive buffer and setback from mine operations. The 313 acre mine pit boundary, developed over the 25 year operation, is centrally located within the 1,911 acre mine and reclamation permit boundary, as displayed in Map 1. Considered from this maximum 25 year mine pit boundary, setbacks from one quarter mile up to three quarters of a mile are provided, and average in excess of one half mile. Minimum one quarter mile setbacks occur at the north and south ends of the Hi-Cal pit. These locations will receive first priority for development of landscaped features adjacent the pit.

7.9.3 Maximum Area of Disturbance

Within the flat valley bottom of Dowe Flats, including the low lying topography to the south along Colorado Highway 66, the landscaping features described above will provide effective visual mitigation of mine operations. As the topographic relief rises at the edges of Dowe Flats along Indian Mountain to the west, Rabbit Mountain on the east, and the Dowe Pass area to the north, the reclaimed out of pit landscape will not provide screening from mine operations. Setbacks provided by the mine permit boundary and the central location of the mine pits within the permit boundary will constrain these unscreened viewpoints to locations at a minimum of one half to one mile distant, depending upon the specific viewpoint. These setbacks will reduce the immediacy of the visual impact, however it is clear that from elevated areas adjacent to Dowe Flats some portions of the mine operations will be visible. Due to the nature of the pit quarries, where mining proceeds downward into the valley floor, there will be some additional, but limited screening of mine operations from the elevated viewpoints.

In order to reduce the scope of these impacts, the applicant proposes to set an average maximum area of disturbance at 100 acres at any one time during the entire 25 year mine operation. The maximum area of disturbance will typically be as follows:

Pit Quarries	50.0 acres
Ongoing Out of Pit Reclamation	40.0 acres
Permanent Haul Road	6.6 acres
Temporary Haul Roads	3.4 acres
Total	100.0 acres

This 100 acre maximum requires a significant reduction of operational disturbance area as currently practiced at the quarries south of the cement plant and at the Larimer County quarry.

7.9.4 Additional Visual Impacts Mitigation

The Dowe Flats project as proposed also provides for additional visual mitigation as follows:

- retirement and reclamation of the majority of the quarry operations south of the cement plant,
- retirement and reclamation of the Larimer County quarry,
- Dowe Flats reclamation designed and bonded in excess of minimum state regulations,
- Haul Road is the least visually intrusive alternative for bringing the mine product to the plant.

8.0 SPECIAL USE FOR MINING

The Colorado Mined Land Reclamation Board Regular 112 Permit application is incorporated into this Special Use Application by reference. It includes the mining and reclamation plans proposed to Boulder County.

The mine plan objectives are:

1. to provide 25 years of raw material reserves for the cement plant

2. to anticipate and accommodate construction of a reservoir in the East Valley of Dowe Flats during or after mining activity.

The reclamation plan has been developed with the following objectives:

1. Create the most diverse wildlife habitat possible; the diverse topography possible resulting from mining makes this objective particularly attractive and feasible. The opportunities to establish woody vegetation in various parts of the post-mining landscape and take advantage of water entering the lower parts of excavated areas will greatly enhance the value of the area to a variety of game and non-game wildlife.

2. Reduce the strong north-south linear nature of the mining disturbance by introducing topographic grain at differing angles and providing the distraction of varying vegetational texture and from. This necessitates establishment of cross-pit topography as well as topography extending away from the mine area to connect visually with natural topographic features, and again distract the viewer from the mining features underlying the area.

3. The proposed vegetation mosaic, that includes species and plant community structures that occur naturally, will represent an extension of native vegetation on areas adjacent to Dowe Flats, including Rabbit Mountain, Indian Mountain, and other foothills of the northern Front Range.

4. The reclamation plan does not impair and accommodates reservoir construction in the East Valley.

Reclamation will proceed at the earliest stages of mining as topography is built from overburden materials, and topsoil is replaced. Earth will be moved from the area to be mined to its final site by use of the mine haulage trucks, whereupon it will be graded to the desired contours by dozers. Topsoil will be hauled to the site in trucks or scrapers and spread to appropriate depths.

Land use after reclamation will be primarily wildlife habitat and will be consistent with similar

uses of adjacent and surrounding lands, both public and private. Steep land to the west on Indian Mountain and to the east on Rabbit Mountain is currently valuable wildlife habitat. Accordingly, the development of structurally diverse topography, water sources, aquatic habitat, and diverse vegetative communities on what had formerly been primarily plowed marginal agricultural land on the project site will enhance the usefulness of the entire area to wildlife.

9.0 BIBLIOGRAPHY

Aquatic and Wetland Consultants, 1987. Wetland Assessment, Dowe Flats and Foothills Reservoir Properties. Report prepared for the Reservoir Cotenancy.

Aquatic and Wetland Consultants, 1990. Wetland Map, December 11, 1990. Blueline map with commentary.

Boos, M.C. and M.F. Boos, 1957. Tectonics of Eastern Flank Foothills of the Front Range, Colorado. American Association of Petroleum Geologists Bulletin, Vol.41, Pages 2603-2676.

Boulder County Board of Commissioners, 1985. Boulder County Comprehensive Plan: Goals, Policies and Maps, -1-. Adopted March 22, 1978 by Boulder County Planning Commissioner: Accepted and Approved April 6, 1978 by Board of County Commissioners: Revised and Updated in 1985; Planning Commission Adoption on November 28, 1984; and Board of County Commissioner's Acceptance and Approval on February 28, 1985. Document on file, Boulder County Planning Commission, Boulder.

Boulder Board of County Commissioners, 1987. Resolution No. 87-104; A Resolution Concerning Rules and Regulations for County Park and Open Space Land (Superseding Resolutions No. 77-24, 81-173, and 86-118).

Boulder County Parks and Open Space and Boulder County Land Use Department, 1986.Boulder County Comprehensive Plan, Environmental Resources Element -3-, Adopted by the Boulder County Planning Commission November 30, 1983: May 28, 1986 (text) July 30, 1986 (text and map). Approved by the Board of County Commissioners December 8, 1983, and November 18, 1986 (text and map). Document on file, Boulder County Parks and Open Space, Boulder.

Brown, A.M. 1986. Modifying Reservoir Fish Habitat with Artificial Structures. Reservoir Fishery Management: Strategies for the 80's. Pages 98-102. American Fisheries Society. Bethesda, MD.

Burney, Michael S. 1989a. A Briefing Paper on American Indian Site 5BL15/18 on the West Side of Dowe Flats, Boulder County, Colorado. September 8th, Paper on file, Burney and Associates, Boulder.

Burney, Michael S. 1989b. A History of Archaeological Investigations and Inventory ofAmerican Indian Resources of Dowe Flats, Including Indian Mountain to the West and Rabbit Mountain to the East, Northern Boulder County, Colorado. September 24th. Report on file, Colorado Historical Society. Office of Archaeology and Historic Preservation Denver.

Burney, Michael S. 1989c. Some Recommendations for an Initial Management Strategy of the Archaeological and Historical Resources Known, and Yet to be Recorded for the Indian Mountain, Dowe Flats, and Rabbit Mountain Area. Northern Boulder County, Colorado. October 13th Report on file, Burney and Associates, Boulder.

Burney, Michael S. 1989d. Dowe Flats Executive Summary, October 14th. Document on file. Burney and Associates, Boulder.

Burney, Michael S. and Jennifer Genner, 1991. An Inventory for American Indian Cultural Resources Located Within Selected Portions of Dowe Flats Between Indian Mountain and Rabbit Mountain East of Lyons, Boulder, County, Northern Colorado. (In Preparation.)

Cassells, E.S., and R.N. Farrington, 1986. Excavations at the Indian Mountain Site. 5BL876: A Multi-Component Stone Circle Site in Colorado's Northeastern Foothills. Southwestern Lore 31(112):129-139.

Colorado Water Resources and Power Development Authority, 1986. St. Vrain Basin Reconnaissance Study, Summary Report.

Cummings, L.S. 1991. Pollen and Phytolith Analysis at 5BL2431: A Stratified Multi-Component Site Near Lyons, Boulder County, Colorado. Document on file, Burney and Associates, Boulder.

ESCO Associates, Inc., 1987. Dowe Flats Limestone Ridge Plant Species Inventory and Vegetation Mapping. Report prepared for the Reservoir Companies, 1707 N. Main, Suite 304, Longmont, Colorado 80501.

Fox Consultants, Inc. 1985. Report on Geotechnical Feasibility and Preliminary Embankment

Design, Proposed Dowe Flats Dam and Reservoir near Lyons, Colorado. Report prepared for St. Vrain Mutual Reservoir and Water Company. 7348 Ute Highway, Longmont, Colorado 80501.

Grant, Marcus P. 1990. Test Excavations at 5BL2431: A Stratified Multicomponent Site Near Lyons, Boulder County, Colorado. Report on file, Burney and Associates, Boulder.

Holistic Resource Management, 1987. Dowe Flats Agricultural Report, Report prepared for Reservoir Cotenancy Venture, 700 Florida, Suite 100, Longmont, Colorado 80501.

Hydroshpere, 1990. CBT/Windy Gap Operations Study, Final Report. Decision Support Document. Prepared for the Municipal Subdistrict, Northern Colorado Water Conservancy District.

Hunter, Z. 1947. Structural Geology of Boulder County. Ph.D. Dissertation, University of Colorado, Boulder.

Lowman, B.M. 1977. Stratigraphy of the Upper Benton and lower Niobrara Formations(Upper Cretaceous) Boulder County, Colorado. Master's Thesis, Colorado School of Mines, Golden.

Mallette, C.M. 1962. Raw Materials Investigation, Dowe Flats, Colorado. Unpublishedreport prepared for Ideal Basic Industries, 23 pp.

Masters, C.D. 1957. Structural Geology of the Rabbit Mountain-Dowe Pass Area. Master's Thesis. University of Colorado, Boulder, 60pp.

Mathews, V. III, D.F. Work, W.E. LeMasurier, and D.W. Stearns, 1976. Mechanism of Laramide Deformation along the Northeastern Flank of the Colorado Front Range. Professional Contributions of Colorado School of Mines, No. 8, Pages 386-397.

Meier, T.J. 1987. The Impact of the Town of Noland on the Indian Mountain/Rabbit Mountain Area. Ms. on file, Burney and Associates, Inc., Boulder.

Moomaw, J.C. 1960. The "Ring Makers." Southwestern Lore 25(4):5-9.

Nykamp, R.H. 1984a. Rabbit Mountain Survey: Letter Progress Report to Boulder County Parks and Open Space. Letter on file, Burney & Associates, Boulder.

Nykamp, R.H. 1984b, Field Notes Typed by Sue Lippincott, June, 1989 for Rabbit Mountain Survey as Recorded by Robert H. NyKamp on February 21 and 25, April 15 and May 19, 1984. Ms. on file, Burney & Associates, Boulder.

Report of the Committee on Water Quality Criteria, 1972. Federal Water Pollution Control Administration, Washington, D.C., 234 pp.

Soil Conservation Service, 1975. Soil Survey of Boulder County Area, Colorado U.S. Department of Agriculture, Soil Conservation Service in cooperation with Colorado Agricultural Experiment Station.

U.S. Fish and Wildlife Service,. 1989. Black-footed ferret survey guidelines for compliance with the Endangered Species Act. USFWS, Denver, Colorado and Albuquerque, New Mexico.

Western Ecosystems Inc., 1987. Preliminary Assessment of Wildlife Resources and Wildlife Issues Associated with Reservoir Development in Dowe Flats, Boulder County, Colorado. Report prepared for Reservoir Cotenancy Venture.

Western Historical Studies, Inc. 1990. Literature Review and Historic Reconnaissance Survey Report, Dowe Flats, Boulder County, Colorado. Report prepared for Western Technologies, Inc., 13949 West Colfax Avenue, Number 140, Golden, Colorado 80401.

Wheeler, Brent, 1989. City of Boulder Open Space Cultural Resources Guidelines (Draft), December 13th. Document on file, Boulder County Parks and Open Space Department, Boulder.

TABLE 5.5.6-1

Archaeology Projects Completed at Dowe Flats between April 18, 1989, and February 3, 1990.

Author and Date	Subject
Burney 1989a	Archaeological survey of the east side of Dowe Flats from the Supply Ditch north to elevation 5400
Burney 1989b	Initial Archaeology Management Strategy
Burney 1989c	Dowe Flats Archaeology Executive Summary
Grant 1990	The limited testing of 5BL2431
Burney and Germer 1991	Dickens/Peicker, Harroun, and Frontier Materials properties
Burney and Germer 1991	The northern section of the West Valley of Dowe Flats reserved for the overburden generated during the first seven years of mining
Burney and Germer 1991 in the West Valley	County Road 47 realignment corridor
Burney and Germer 1991	Portions of the Card property relevant to the proposed County Road 47 realignment corridor
Burney and Germer 1991 Ridge	Mine Haul Road in the "incidental use area" west of the Limestone
Burney and Germer 1991 the 5400' pool elevation	Northern portions of the East Valley and Limestone Ridge above
Burney and Germer 1991	Reconnaissance of site 5BL793
Cummings 1991	Pollen and phytolith analysis at 5BL2431

TABLE 5.5.6-2

Proposed Cultural Resources Management Process

1. Submit all cultural resources reports to State Historical Preservation Office (SHPO) for review, (completed March, 1993).

2. Introductory meeting with SHPO, Boulder County Parks and Open Space Dept. (BCPOS) and project consultants to review submitted reports and discuss overall project design (completed May, 1993).

3. Follow-up meeting with SHPO and project consultants to develop a Cultural Resources Management Plan (including a draft mitigation plan, Indian consultation process, paleontology, historic archaeology, and a review of the ethnographic, ethnohistoric, and historic literature for Indian use of the Dowe Flats area, including Lyons and the St. Vrain Creek).

4. Submit Special Use Permit Application and Site Specific Development Plan.

5. Complete Cultural Resources Management Plan, possibly including a Memorandum of Agreement (MOA) with SHPO and BCPOS.

6. Review MOA with Lyons Redstone Museum and the Colorado Archaeological Society. Define the role these organizations will fill in the execution of the Cultural Resources Management Plan.

7. Incorporate applicable terms of MOA and Cultural Resources Management Plan into the Boulder County Special Use Permit and Site Specific Development Plan for the mine and reclamation plan.

8. Interim reviews of the Cultural Resources Management Plan, as provided in the terms of the Site Specific Development Plan and/or Development Agreement.